CORE MATHEMATICS GRADE 10: SURFACE AREA AND VOLUME OF SOLIDS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.7 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.7: Surface Area and Volume of Solids
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Surface area of prisms (triangular, rectangular, and other prisms) – Surface area of pyramids (square-based and other pyramids) – Surface area of cones – Surface area of frustums (cone frustums) – Surface area of spheres – Surface area of composite solids – Volume of prisms (triangular, rectangular, and other prisms) – Volume of pyramids (square-based and other pyramids) – Volume of cones – Volume of frustums – Volume of spheres – Volume of composite solids – Real-life applications of surface area and volume of solids
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) determine the surface area of prisms, pyramids, cones, frustums and spheres, b) determine the surface area of composite solids, c) determine the volume of prisms, pyramids, cones, frustums and spheres, d) determine the volume of composite solids, e) apply surface area and volume of solids to real-life situations, f) appreciate the use of surface area and volume of solids in real-life situations.
Core Competencies	– Communication and Collaboration: the learner discusses and shares findings with peers when measuring and calculating the surface area and volume of locally available solid models such as tins, packets and containers used for ugali flour or cooking oil. – Critical Thinking and Problem Solving: the learner analyses composite solid shapes found in real-life Kenyan contexts — such as grain silos in Eldoret or water tanks in rural Rift Valley — and decomposes them to determine total surface area and volume. – Creativity and Imagination: the learner improvises and constructs nets and models of prisms, pyramids, cones and spheres

	using locally available materials (cardboard, manila paper) to verify surface area formulae. – Digital Literacy: the learner uses digital devices and software (where available) to explore 3D solid models, nets and volume visualisations. – Self-Efficacy: the learner builds confidence in applying surface area and volume formulae independently to solve real-world packaging, construction and storage problems relevant to Kenyan industries.
Core Values	– Responsibility: the learner carefully handles models of solids and measurement tools collected from the immediate environment, and accurately records measurements to ensure correct calculations of surface area and volume. – Integrity: the learner applies formulae with accurate measurements and honest working, adhering to mathematical rigour when solving surface area and volume problems. – Unity: the learner works collaboratively in groups to construct solid models, measure dimensions and verify calculated surface areas and volumes, appreciating each member's contribution. – Respect: the learner recognises and values the input of every group member during practical activities involving the construction and measurement of solid models.
Science & Engineering Practices	– Using Mathematics and Computational Thinking: learners apply and manipulate formulae for surface area and volume of prisms, pyramids, cones, frustums, spheres and composite solids, using arithmetic and algebraic reasoning to solve quantitative problems. – Planning and Carrying Out Investigations: learners plan and conduct hands-on measurement investigations using real solid objects (tins, boxes, cones) to collect dimension data, then compute and compare calculated surface area and volume against estimates. – Analysing and Interpreting Data: learners examine measured dimensions and computed results, identify errors or inconsistencies, and draw conclusions about which solid shape optimises volume for a given surface area — relevant to packaging and storage design. – Constructing Explanations: learners construct clear mathematical explanations of how surface area and volume formulae are derived from nets and cross-sectional areas, and explain results in context (e.g. why a cylindrical water tank in Kisumu holds more than a box of the same surface area). – Obtaining, Evaluating, and Communicating Information: learners present their findings on surface area and volume of composite solids to the class, evaluating the reasonableness of results and communicating solution strategies clearly.
Pertinent & Contemporary Issues (PCIs)	– Education for Sustainable Development (Environmental Education): the learner uses and repurposes locally available materials such as cardboard packaging, tin cans and plastic containers to construct solid models, promoting environmental consciousness and waste reduction. – Social Cohesion: the learner works in groups to build models, measure solids and solve problems, fostering teamwork and mutual respect across diverse learner backgrounds. – Socio-Economic Issues: the learner applies surface area and volume concepts to real Kenyan economic contexts such as determining the amount of metal sheet needed to fabricate a water tank in Nakuru, or the capacity of a grain store in the Rift Valley, linking mathematics to livelihoods and national development. – Home Science and Technology: the learner connects volume calculations to everyday household decisions such as estimating the volume of a sufuria used for cooking ugali or the capacity of a storage jerrycan.
Career Connections	– Architecture and Civil Engineering: understanding surface area informs material estimation for roofing, wall cladding and flooring of buildings — directly relevant to Kenya's affordable housing programmes. – Manufacturing and Packaging Industry: volume and surface area calculations are essential for designing and optimising packaging for Kenyan food products such as maize flour, cooking oil and tea from Kericho. – Water and Sanitation Engineering: volume of cylinders, frustums and spheres is applied in designing water storage tanks,

	pipes and reservoirs for communities around Lake Victoria and arid regions. – Agriculture and Agribusiness: farmers in the Rift Valley and Central Kenya use volume concepts to estimate grain storage capacity in cylindrical silos and conical heaps. – Medicine and Pharmacy: volume of spherical capsules and cylindrical vials is critical in drug dosage calculations and medical device manufacturing. – Surveying and Land Management: surface area concepts underpin calculations of land volumes in road construction and mining operations in regions such as Taita Hills.
Focus for Lessons	This unit develops learners' ability to calculate the surface area and volume of common solids — prisms, pyramids, cones, frustums, spheres and composite solids — grounding every formula in the physical reality of objects found in Kenyan daily life (grain tins, water tanks, roofing cones, sports balls). Learners move from hands-on net construction and measurement of real objects to formal derivation and application of formulae, building both procedural fluency and conceptual understanding. The unit culminates in a real-life project where learners design or evaluate a practical solid structure (e.g. a water storage tank or grain silo) relevant to their local community, integrating surface area and volume to make informed decisions.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction? KICD KEY INQUIRY QUESTIONS: 1. How do we determine the surface area and volume of solids? 2. How are surface area and volume of solids applied in real-life situations?
Anchoring Phenomenon	A local posho mill owner in Eldoret has two large metal storage containers for holding maize grain: one is a tall cylindrical drum and the other is a rectangular metal box. Both containers appear roughly the same size from the outside, and the metalsmith who made them used the same total area of iron sheet for each. Yet when grain is poured in, the cylindrical drum holds noticeably more maize than the rectangular box. The mill owner is puzzled — if the same amount of metal was used for both, why does one hold more grain? Students observe photographs and physical models of both containers, measure their dimensions, and are challenged to explain the discrepancy. This phenomenon is puzzling because learners intuitively assume equal surface area means equal volume, but discover that shape — not just material used — determines storage capacity, launching an investigation into how surface area and volume are related yet distinct properties of three-dimensional solids.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Surface Area of Prisms: Learners observe the anchoring phenomenon (two containers, same metal sheet, different grain capacity). They predict which holds more and why. They construct nets of rectangular and triangular prisms from cardboard, measure faces, and derive the surface area formula. They calculate the surface area of the rectangular box from the phenomenon and record findings on the Driving Question Board (DQB). Lesson 2 — Surface Area of Pyramids: Learners examine pyramid-roofed structures common in Kenyan homesteads (e.g. traditional Maasai manyatta roofs and school hall rooftops). They construct nets of square-based and triangular pyramids, identify slant height versus vertical height, and derive the surface area formula. They solve problems involving material needed to cover a pyramid-shaped roof. Lesson 3 — Surface Area of Cones and Frustums: Learners investigate the cone shape using real objects (ice-cream cones, conical grain heaps, funnel tops of water tanks). They derive the curved surface area formula using the net of a cone (a sector of a circle). They extend this to the frustum by considering a cone with its top removed, linking to the Turkana water-tank supporting phenomenon. Learners calculate the metal sheet needed for a conical tank top.

	Lesson 4 — Surface Area of Spheres and Composite Solids: Learners explore the surface area of spheres by wrapping a ball in paper strips, then verify using the formula $SA = 4\pi r^2$. They extend to composite solids (cylinder + cone; hemisphere + cylinder) and practise decomposing shapes. They connect to the Nairobi packaging factory supporting phenomenon and the football supporting phenomenon. The DQB is updated with new understanding. Lesson 5 — Volume of Prisms and Pyramids: Learners return to the anchoring phenomenon and now investigate volume. They fill rectangular and triangular prism models with sand or water and compare. They derive Volume = base area × height for prisms and $V = \frac{1}{3} \times$ base area × height for pyramids by comparing filled models. They compute the volume of the rectangular box from Lesson 1 and compare it with the cylinder (previewing Lesson 6). Lesson 6 — Volume of Cones, Frustums and Spheres: Learners use the relationship between cone and cylinder volumes ($V_{cone} = \frac{1}{3} V_{cylinder}$) demonstrated by pouring water between matching models. They derive $V = \frac{1}{3}\pi r^2 h$ for cones and $V = 4/3\pi r^3$ for spheres. They calculate the volume of the cylindrical drum from the anchoring phenomenon and formally resolve why it holds more grain than the rectangular box of equal surface area. Lesson 7 — Volume and Surface Area of Composite Solids & Real-Life Applications: Learners tackle multi-step problems involving composite solids such as a cylindrical water tank with a hemispherical top (common in Kenyan water projects). They calculate both the total surface area (painting/material cost) and volume (water storage capacity). They also work on a grain silo problem set in the Rift Valley, integrating all formulae learned. Lesson 8 — Project Presentations, Model Building & Unit Consolidation: Learner groups present their home/extended project: designing a storage container (grain silo, water tank or cooking pot) for a Kenyan community need, specifying dimensions, total surface area of material required and volume/capacity. Groups critique each other's designs. The class builds a final cumulative model on the DQB linking surface area and volume formulae to the anchoring phenomenon, resolving the driving question fully and discussing careers that use these concepts.
Supporting Phenomena	– A mandazi vendor in Mombasa notices that ball-shaped (spherical) mandazi cool down more slowly than flat disc-shaped ones of the same mass — prompting investigation into how surface area relative to volume affects heat loss and connects to the surface area of spheres. – A water harvesting project in Turkana County must choose between a conical, cylindrical and hemispherical roof-catchment tank; all use the same amount of plastic sheeting — students investigate which shape stores the most water, exploring volume and surface area of cones, cylinders and spheres. – Workers at a Nairobi packaging factory notice that when a conical top is placed on a cylindrical can (like a composite solid), the total surface area to be painted changes in a non-obvious way — prompting learners to decompose composite solids and calculate total surface areas systematically. – A school in Kisumu receives a donation of spherical footballs and asks: how much leather (surface area) is needed to make each ball, and how much air (volume) does it hold? — anchoring the surface area and volume of spheres in a familiar, motivating context.

LESSON 1 (80 minutes): Anchoring Phenomenon & Surface Area of Prisms: Same Metal Sheet, Different Grain Capacity?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce the anchoring phenomenon of two metal storage containers made from the same iron sheet but holding different amounts of maize grain, and to derive the surface area formula for rectangular and triangular prisms by constructing and measuring their nets.
Knowledge	Learners will know that: (i) surface area is the total area of all faces of a three-dimensional solid; (ii) the surface area of a rectangular prism is $SA = 2(lw + lh + wh)$; (iii) the surface area of a triangular prism is $SA = 2(\frac{1}{2}bh_t)$ + (perimeter of triangle × length of prism); (iv) equal surface area does NOT imply equal volume — shape determines storage capacity.
Skills	Learners will be able to: (i) construct and measure the net of a rectangular prism and a triangular prism from cardboard; (ii) identify and label all faces of a prism; (iii) calculate the surface area of a rectangular prism using the formula; (iv) record structured observations and calculations on a Driving Question Board (DQB).
Attitudes	Learners will appreciate that: (i) mathematics is a practical tool used by Kenyan artisans, metalworkers, and posho mill owners to make cost-effective and efficient design decisions; (ii) questioning intuitive assumptions — such as 'same material = same capacity' — is the foundation of mathematical inquiry; (iii) collaborative investigation and honest recording of predictions and findings builds integrity and scientific thinking.
Key Inquiry Question	How do we determine the surface area of solids, and why does the same area of metal sheet produce containers that hold different amounts of maize grain?
Purpose in Storyline	This is the opening lesson of the sub-strand. It launches the anchoring phenomenon — two metal containers at a posho mill in Eldoret, built from equal areas of iron sheet, yet holding different volumes of maize grain. Learners make predictions, construct prism nets, derive the surface area formula for rectangular prisms, and calculate the surface area of the rectangular container from the phenomenon. The unresolved puzzle (why does the cylinder hold more?) is posted on the DQB, anchoring all future lessons in a real Kenyan problem.
Safety Notes	When using scissors or craft knives to cut cardboard nets, learners must cut away from the body and place sharp tools face-down on the desk when not in use. Rulers used as cutting guides must be held firmly. No horseplay with tools. Teacher circulates continuously during the net-construction activity.

B. LESSON OVERVIEW

Learners open the sub-strand by encountering a puzzling real-world phenomenon: a posho mill owner in Eldoret has two metal storage containers — a cylindrical drum and a rectangular metal box — both made from the same total area of iron sheet. Yet the cylinder holds noticeably more maize grain than the box. In Phase 1 (Predict), learners examine photographs and physical models of both containers, record individual written predictions, and share reasoning in a Think-Pair-Share. In Phase 2 (Observe), learner groups construct cardboard nets of rectangular and triangular prisms, measure all faces with rulers, and record face areas in a structured table. In Phase 3 (Explain), the class derives the surface area formula for rectangular and triangular prisms from their net data, and calculates the surface area of the rectangular box from the phenomenon using given dimensions. In Phase 4 (DQB Creation), learners post confirmed findings and unresolved questions — especially why the cylinder holds more — onto the class Driving Question Board. In Phase 5 (Model Building), learners sketch a labelled net of the rectangular box from the phenomenon and annotate it with face areas, creating the first page of their cumulative geometric model portfolio.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Volume of triangular prism & cube] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners examine two photographs displayed on the board (or physical cardboard models): a tall cylindrical drum and a rectangular metal box, both labelled as storage containers from Mama Wanjiru's posho mill in Eldoret. The teacher reads aloud the scenario: 'The metalsmith used exactly the same area of iron sheet for both containers. When Mama Wanjiru fills them with maize grain, the cylindrical drum holds more. How is this possible?' Each learner silently writes a prediction on a sticky note (1 minute), answering: (a) Which container do you think holds more — and did you already know that before reading? (b) Why do you think equal sheet area does NOT mean equal grain capacity? Learners then share predictions in pairs (1 minute each), before 4–5 pairs are cold-called to share with the class. Predictions are displayed on a designated 'Our Predictions' section of the board — NOT erased — to be revisited at the end of the lesson.	1. Display the two container photographs side by side on the board. Point to each: 'This is a cylindrical drum. This is a rectangular metal box. Both were made by the same fundi in Eldoret using the same total area of iron sheet.' Pause 10 seconds — allow silence. 2. Ask: 'On your sticky note, write your prediction: which holds more grain, AND why do you think equal surface area does not mean equal volume? You have 60 seconds — write silently.' Set a visible 60-second timer. 3. After writing, say: 'Turn to your partner — share your prediction and listen to theirs. You have 90 seconds.' 4. After pairs share, cold-call 5 pairs (select a mix of confident and quieter learners): 'Baraka, what did you and your partner predict — and what was your reasoning?' Apply 10-second WAIT TIME after each response before probing. 5. After each response, ask the class: 'Raise your hand if your prediction agreed with theirs.' Do NOT confirm or deny any prediction yet. Say: 'We will find out if your predictions were correct by the end of this lesson — and we may still have questions that take us several more lessons to fully answer.' 6. Post all sticky notes on the board under the heading 'OUR PREDICTIONS — DO NOT ERASE.'	Think-Pair-Share with Anchored Prediction Posting. This strategy activates prior knowledge about area and capacity, surfaces misconceptions (especially the intuitive but incorrect belief that equal surface area implies equal volume), and creates cognitive dissonance that motivates the investigation. By physically posting — not erasing — predictions, learners are held accountable to their prior thinking and can track how their understanding evolves across the unit.	Observable indicator: Every learner has a sticky note with a written prediction before pairs share. Teacher scans sticky notes during pair-share for the following diagnostic signals — (a) learners who correctly predict the cylinder holds more but cannot explain why (surface-area/volume distinction not yet formed); (b) learners who predict the box holds more (strong misconception to address); (c) learners who correctly invoke 'shape matters' even without formal vocabulary. Teacher makes a mental note of category (b) learners for targeted questioning during Phase 3.
Observe Phase 📹 **VIDEO:** [Volume of triangular prism &	In groups of 4, learners receive a pre-printed net template on A3 cardboard (one rectangular prism net and one triangular prism net per group), scissors, rulers, and a	1. Distribute materials. Say: 'Your cardboard net is like the flat iron sheet the Eldoret fundi starts with. Your job is to find the total area of that sheet — that is what we call	Concrete-to-Abstract Net Construction. By physically cutting, folding, measuring, and summing face areas BEFORE seeing the formula, learners build a concrete	Observable indicators during circulation: (a) All group members are measuring, not just one learner — teacher redirects if one learner is dominating. (b) Learners

cube Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	structured observation table (printed worksheet). Step 1: Learners carefully cut out the rectangular prism net, fold it WITHOUT gluing, and identify each face. Step 2: Using a ruler, they measure the length and width of each face and record in the table: Face Name | Shape | Length (cm) | Width (cm) | Area (cm²). Step 3: They repeat for the triangular prism net, carefully measuring the triangular faces (base and height) and the three rectangular faces. Step 4: Each group calculates the area of every individual face and records the total (sum of all face areas) in the last row of their table. Step 5: Groups compare their total face areas across groups to verify consistency. The teacher connects this explicitly to the phenomenon: 'The iron sheet the fundi in Eldoret used is like this net — it is the flat material that gets folded into the 3D container. The total area of your net IS the surface area of the container.'	surface area.' 2. Circulate during cutting. After 3 minutes, stop the class: 'Freeze — show me your net before folding. Count the number of faces on your rectangular prism net. Tell your partner.' Cold-call 3 groups: 'Akinyi's group — how many faces? What shape is each face?' Apply 10-second WAIT TIME. 3. After folding: 'Without measuring yet — look at the rectangular prism. Which faces look identical to each other? Discuss in your group for 30 seconds.' Cold-call 2 groups. Expected answer: three pairs of identical rectangles. Affirm: 'Good — notice that. We will use that observation when we write the formula.' 4. During measurement: circulate and check that learners are measuring length AND width (not just one dimension). Prompt: 'You have measured the length — what else do you need to find the area of a rectangle?' 5. After totals are calculated, ask the class: 'Group totals — call them out.' Write all group totals on the board. Ask: 'Why might totals differ slightly across groups?' (Measurement precision, rounding.) Say: 'This is why engineers and fundis must measure very carefully.' 6. Explicitly link back to phenomenon: 'The number you just calculated — the total face area — is exactly what the Eldoret metalsmith measured out in iron sheet. Now we need to ask: does the cylinder's net give the same total area? We will find out in later lessons.'	referent for what 'surface area' means. The net makes the abstract concept of 'total outer area of a 3D solid' visible and touchable. Connecting the net explicitly to the iron sheet in the phenomenon grounds the mathematics in a real Kenyan artefact — the storage container — rather than an abstract shape.	correctly calculate area as length × width for rectangular faces and ½ × base × height for triangular faces — teacher checks at least 3 groups' triangular face calculations. (c) Group totals for the same net template should agree within ±2 cm² — groups with large discrepancies are asked to re-measure. Exit check: teacher asks one group to point to and name all 6 faces of the rectangular prism before they proceed to Phase 3.
Explain Phase 📹 **VIDEO:** Volume of	The class reconvenes for whole-class sensemaking. The teacher leads a structured discussion to	1. Draw a large net of a rectangular prism on the board. Label faces Top, Bottom, Front,	Collaborative Formula Derivation (Structured Inquiry). Rather than stating the formula, the teacher	Observable indicators: (a) During formula derivation, at least 3 different learners contribute a step

triangular prism & cube Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	derive the surface area formula for a rectangular prism from the group's own net data. Learners first share which faces they found to be identical (three pairs). The teacher writes on the board: SA = Area of Face 1 + Area of Face 2 + ... and guides learners to group the pairs: SA = 2(lw) + 2(lh) + 2(wh) = 2(lw + lh + wh). Learners copy the derived formula into their exercise books with a worked example. They then apply the formula directly to the anchoring phenomenon: the teacher announces the dimensions of the rectangular metal box from the Eldoret posho mill — length = 120 cm, width = 80 cm, height = 100 cm — and learners individually calculate its surface area. They compare their answer with a partner. The triangular prism formula is similarly derived: SA = $2(½bh_t) + (a + b + c) × L$, where a, b, c are the sides of the triangular face and L is the length of the prism. The teacher uses a real-world application: a triangular metal water trough used by Maasai pastoralists in the Rift Valley. Learners solve one practice problem individually.	Back, Left, Right. Ask: 'From your group work — which faces were identical? Call out the pairs.' Cold-call 3 groups. Record on the board. 2. Say: 'So surface area = sum of all six face areas. Can we write this more efficiently using what we know about identical pairs?' Wait 15 seconds. Cold-call one learner: 'Njoroge — try to write a shorter formula using the pairs.' Build the formula on the board collaboratively. 3. State clearly: 'The formula we have derived together is: SA = 2(lw + lh + wh). Write this in your books with a box around it.' 4. Announce the phenomenon dimensions: 'The rectangular metal box from Mama Wanjiru's posho mill in Eldoret has length 120 cm, width 80 cm, height 100 cm. Using our formula, calculate its surface area. You have 3 minutes — work individually in silence.' Set timer. 5. After 3 minutes: 'Check your answer with your partner.' Then cold-call one pair: 'Fatuma and Otieno — what did you get? Walk us through your substitution step by step.' Expected: SA = 2(120×80 + 120×100 + 80×100) = 2(9600 + 12000 + 8000) = 2(29600) = 59,200 cm². Write full solution on the board. 6. Ask: 'So the Eldoret metalsmith used 59,200 cm² of iron sheet for the box. Now — if the cylinder was also made from 59,200 cm² of iron sheet, what dimensions might it have? We cannot fully answer that yet. Post that question — it is going on our DQB.' 7. Derive triangular prism formula using the same net-grouping approach. Ask: 'What is different about the triangular prism? How many rectangular faces? How many triangular	guides learners to construct it from their own net measurements, making the formula a logical consequence of grouping identical faces rather than an arbitrary rule to memorise. This builds conceptual understanding (WHY the formula has the structure it does) alongside procedural fluency. Anchoring the calculation to the exact dimensions of the Eldoret rectangular box maintains the phenomenon connection and gives the surface area a real, meaningful value (59,200 cm² of iron sheet).	— teacher tracks participation by marking names. (b) Individual calculation: teacher circulates during the 3-minute silent work and checks that learners correctly substitute l=120, w=80, h=100 — common error is using perimeter instead of area for faces. (c) Cold-called pair must show full working (substitution → simplification → answer) on the board — teacher corrects any arithmetic error immediately and asks the class: 'Do you agree? Show thumbs up/down.' (d) Teacher asks one learner to state the formula from memory without looking at their book — a minimum of 2 learners should be able to do this by end of phase.

		faces?' Wait 10 seconds before cold-calling.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Volume of triangular prism & cube Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class creates the unit's Driving Question Board together. The teacher draws (or reveals a pre-prepared) large board divided into four columns: (1) WHAT WE OBSERVED, (2) WHAT WE LEARNED TODAY, (3) QUESTIONS WE STILL HAVE, (4) HOW THIS CONNECTS TO THE PHENOMENON. Each learner receives two sticky notes. On the first, they write one thing they observed or learned today. On the second, they write one question they still cannot answer. Learners post their notes in silence, then the teacher reads out a selection, grouping recurring questions. The teacher then writes the unit's DRIVING QUESTION at the top of the board in large letters: 'Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?' Specific unresolved questions from learners (e.g., 'What is the surface area formula for a cylinder?', 'Why does shape affect how much it holds?', 'Can we find the dimensions of the cylindrical drum?') are posted under Column 3 and flagged with a star to indicate they will be answered in upcoming lessons.	1. Reveal the pre-prepared DQB (large chart paper or section of the whiteboard divided into 4 columns). Explain: 'This board will track our growing understanding across every lesson in this sub-strand. It stays on the wall for the whole unit.' 2. Distribute two sticky notes per learner. Say: 'On your YELLOW note, write one thing you observed or learned today — be specific, use numbers if you can. On your PINK note, write one question you cannot yet answer. You have 90 seconds.' 3. Learners post notes silently. Teacher reads 4–5 from each column aloud. Cluster similar notes. 4. Read out and write the Driving Question prominently: 'Here is the big question that will guide our entire investigation: WHY does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?' 5. Point to the phenomenon photographs still on the board: 'Everything we do in this sub-strand connects back to Mama Wanjiru's posho mill in Eldoret. Every formula we derive is a tool to help answer this question.' 6. Star the most important unresolved questions: 'These starred questions — especially about the cylinder — we cannot answer yet. Watch for when we CAN answer them and come back to update this board.' 7. Ask the class: 'What shape do we need to study next to get closer to answering the driving question?' Expected: cylinder. Affirm: 'Correct	Driving Question Board as a Living Knowledge Tracker. The DQB externalises the class's collective knowledge and uncertainty, making thinking visible. It creates a sense of purposeful incompleteness — learners can see exactly what they do not yet know — which motivates future lessons. By physically posting and reading out questions, the teacher validates curiosity as a core mathematical practice. The four-column structure (Observed / Learned / Questions / Phenomenon Connection) mirrors the scientific practice of distinguishing evidence from inference.	Observable indicators: (a) Every learner posts at least one 'Learned' note and one 'Question' note — teacher scans to ensure no blank notes. (b) Quality check on 'Learned' notes: notes should reference specific mathematical content (e.g., 'SA of the rectangular box = 59,200 cm²', 'SA = 2(lw + lh + wh)') rather than vague statements ('I learned about shapes'). Teacher points out two strong examples and reads them to the class as models. (c) Quality check on 'Question' notes: questions should be answerable through mathematics (e.g., 'What is the surface area of a cylinder?') — teacher redirects off-topic questions. (d) The driving question is written verbatim and visible to all learners.

		— the cylinder is coming.'		
Model Building Phase 📹 **VIDEO:** [Volume of triangular prism & cube] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners begin their cumulative Geometric Model Portfolio — a notebook or set of pages that will be updated every lesson with revised diagrams and annotations as their understanding grows. In this first entry, each learner draws a neat, to-scale (or clearly proportioned) net of the rectangular metal box from the Eldoret posho mill (l = 120 cm, w = 80 cm, h = 100 cm). They label every face with its name (Top, Bottom, Front, Back, Left Side, Right Side), its dimensions, and its individual area in cm². They then write the total surface area calculation beneath the net. In the 'Model Reflection' box at the bottom of the page, they respond to the prompt: 'This net represents the iron sheet the Eldoret fundi cut out. What does this model NOT yet tell us about how much grain the box can hold?' This written reflection surfaces the distinction between surface area and volume, which is the core conceptual tension of the entire sub-strand.	1. Introduce the portfolio: 'Open to a fresh page in your exercise book — or use the portfolio sheet I am handing out. This is the first page of your Geometric Model Portfolio. Every lesson, we will update this with new diagrams and understanding.' 2. Say: 'Draw the net of the rectangular box from our phenomenon. Use the dimensions: length 120 cm, width 80 cm, height 100 cm. Label every face — its name, dimensions, and area. You have 8 minutes.' Set timer. Circulate and check: (a) all 6 faces are present on the net, (b) dimensions are labelled correctly on each face, (c) areas are calculated and written on each face. 3. After 8 minutes: 'Beneath your net, write the full surface area calculation using the formula SA = 2(lw + lh + wh). Show every substitution step.' 4. After calculations are written: 'Now answer the Model Reflection prompt in the box at the bottom of your page: What does this net NOT yet tell us about how much grain this box holds?' Give 2 minutes silent writing. 5. Cold-call 3 learners to read their reflection. Expected responses should include ideas like: 'The net shows the outside surface only', 'It does not tell us the space inside', 'We need volume, not surface area, to know how much grain fits.' Affirm: 'Excellent — you have just identified the next big idea we need. Surface area is the outside. Volume is the inside. The Eldoret puzzle is really about the difference between the two.' 6. Close: 'Keep this portfolio page safe. Next lesson, we will add the	Annotated Diagram as Cumulative Model. The Geometric Model Portfolio uses the modelling practice (NGSS SEP 2) to make mathematical understanding tangible and progressive. By annotating each face of the net with dimensions AND areas, learners connect the abstract formula to specific values from the phenomenon. The Model Reflection prompt is carefully designed to force learners to confront the surface-area/volume distinction before it is formally taught — planting a productive question that the next lesson will address. This mirrors how engineers and fundis actually think: drawing, labelling, calculating, and then asking 'what does this drawing NOT yet tell me?'	Observable indicators: (a) Net must show all 6 faces with correct relative positioning — teacher checks at least 6 individual portfolios during the 8-minute drawing time. (b) All face areas must be numerically correct: Top/Bottom = 120×80 = 9,600 cm²; Front/Back = 120×100 = 12,000 cm²; Left/Right = 80×100 = 8,000 cm². Teacher flags any arithmetic errors. (c) Total SA = 59,200 cm² must appear at the bottom with full working. (d) Model Reflection: a strong response must explicitly use the word 'volume' OR reference 'the space inside' — this is the key conceptual target. Teacher identifies 2–3 strong reflections to be read aloud as models. Any learner who writes only 'I do not know' in the reflection box is prompted: 'The net shows the outside of the box — what does it NOT show?'

		surface area of cylinders and get one step closer to solving Mama Wanjiru's puzzle.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did the anchoring phenomenon create genuine cognitive dissonance — did learners seem puzzled and motivated, or did they disengage after the prediction phase? If engagement dropped, consider bringing in a physical model of both containers next lesson. (2) During net construction, which groups struggled most with calculating the area of the triangular face (½ × base × height)? These groups will need additional support in Phase 3 of the next lesson when the triangular prism formula is consolidated. (3) Review the 'Learned' sticky notes from the DQB: how many learners used the formula SA = 2(lw + lh + wh) correctly versus how many wrote vague statements? This reveals the proportion of the class that achieved procedural fluency today. (4) Review the 'Question' sticky notes: are learners asking questions about cylinders and volume — or are they still confused about surface area of prisms? Adjust the opening of Lesson 2 accordingly. (5) Did the Model Reflection prompt surface the surface-area/volume distinction? If fewer than half the class mentioned 'volume' or 'inside the box', open Lesson 2 with a direct discussion of this distinction before introducing cylinders. (6) Note which learners were NOT cold-called today and ensure they are prioritised in Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed photographs and physical models of two metal storage containers from a posho mill in Eldoret — a cylindrical drum and a rectangular metal box — both made from the same total area of iron sheet. They also observed the constructed cardboard nets of rectangular and triangular prisms and measured the area of every individual face.
What did I learn?	Learners learned that: (1) surface area is the total area of all faces of a three-dimensional solid, calculated by summing the areas of every face in the net; (2) the surface area formula for a rectangular prism is SA = 2(lw + lh + wh); (3) the surface area formula for a triangular prism is SA = 2(½bh_t) + (perimeter of triangular face × length of prism); (4) the rectangular metal box from the Eldoret posho mill (l=120 cm, w=80 cm, h=100 cm) has a surface area of 59,200 cm², representing the total iron sheet used by the fundi.
How does this explain the phenomenon?	Learners explained — partially — why a net represents the flat material used to construct a 3D container, and began to articulate that surface area (the iron sheet used) does not determine volume (how much grain fits inside). This explanation is incomplete: learners have posted the unresolved question 'Why does the cylinder hold more if the same sheet was used?' on the DQB, which drives the remainder of the sub-strand investigation.

LESSON 2 (80 minutes): Surface Area of Pyramids: Designing Pyramid Roofs for Kenyan Structures	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners derive and apply the surface area formula for square-based and triangular pyramids by constructing nets, distinguishing slant height from vertical height, and solving real-life problems involving material estimation for pyramid-shaped roofs common in Kenyan homesteads and school buildings.
Knowledge	By the end of the lesson, the learner should be able to: (a) identify the faces, edges, and vertices of square-based and triangular pyramids; (b) distinguish between slant height and vertical height of a pyramid; (c) construct and interpret nets of pyramids; (d) derive the formula for the surface area of a pyramid (SA = base area + ½ × perimeter of base × slant height); (e) calculate the surface area of square-based and triangular pyramids.
Skills	By the end of the lesson, the learner should be able to: (a) construct accurate nets of square-based and triangular pyramids using a ruler, pencil, and card; (b) measure slant height and vertical height on physical models and nets; (c) apply the surface area formula to calculate the amount of iron sheet, thatch, or roofing material needed to cover a pyramid-shaped roof; (d) solve multi-step surface area problems in real-life Kenyan contexts.
Attitudes	By the end of the lesson, the learner should: (a) appreciate how mathematical formulas are derived from physical reasoning rather than memorised in isolation; (b) show curiosity about the relationship between the shape of a pyramid's net and its surface area; (c) value the contribution of mathematics to practical trades such as metalsmithing, carpentry, and roofing in Kenyan communities.
Key Inquiry Question	How do we determine the surface area of solids — and how is this applied in real-life situations?
Purpose in Storyline	In Lesson 1 learners established that the cylindrical drum and the rectangular box were built from the same total area of iron sheet (surface area) yet hold different volumes of maize. They now interrogate another real-world surface — the pyramid roof. By computing exactly how much iron sheet or thatch is needed to cover a pyramid-shaped roof, learners deepen their understanding that surface area measures the material covering a solid, not the space inside it. This distinction is critical to resolving the posho mill owner's puzzle: knowing surface area alone does not tell you how much grain a container holds.
Safety Notes	Learners use scissors and craft knives to cut nets from card. Teacher should remind learners to cut away from the body, keep fingers clear of blades, and store cutting tools flat on the desk when not in use. Supervise closely during the cutting activity.

B. LESSON OVERVIEW

Learners examine photographs and physical models of pyramid-roofed structures found across Kenya — traditional Maasai manyatta huts near Narok, the thatched bandas of Malindi, and the corrugated-iron school hall rooftops of Eldoret. They connect these pyramid roofs to the posho mill owner's question: the metalsmith who built the storage containers uses the same iron-sheet-cutting skill to build roofs. Learners predict how much sheet metal a pyramid roof needs, then physically construct nets of square-based and triangular pyramids from card, discover the difference between slant height and vertical height through measurement, derive the surface area formula collaboratively, and apply it to calculate roofing material for pyramid structures. A Driving Question Board entry and cumulative model revision close the lesson.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Volume of a Pyramid - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners examine two photographs displayed at the front: (1) a square-based corrugated-iron pyramid roof on a school hall in Eldoret, labelled with base side = 6 m and a vertical pole of 4 m at the centre; (2) a triangular-based thatched pyramid roof on a Maasai manyatta near Narok. In pairs, learners record predictions in their exercise books: 'How many square metres of iron sheet do you think the school hall pyramid roof needs? How did you estimate?' Each pair writes one sentence linking their prediction to the posho mill owner's phenomenon: 'Is finding the roof area the same kind of problem as finding out how much iron sheet made the drum or the box?' After 4 minutes of pair discussion, the teacher cold-calls 4 pairs to share predictions (selecting a mix of high and low estimates). Predictions are recorded on the chalkboard without evaluation.	Display photographs using a projector or large printed A2 posters (for low-tech settings). Say: 'Look carefully at these two roofs. Both are pyramid shapes — one is iron sheet, one is thatch. The metalsmith in Eldoret who built the grain containers at the posho mill also builds roofs like this. Before we calculate anything, write your best estimate of how much sheet metal the school hall roof needs — and explain your reasoning.' Allow 4 minutes silent/pair thinking (WAIT TIME). Cold-call exactly 4 pairs: 'Pair at the window, what did you predict — and why?' After 4 responses, ask the class: 'Does anyone predict a VERY different number from what we have heard? Why?' Record all estimates on the board in a column labelled OUR PREDICTIONS. Do NOT correct yet. Bridge to observe phase: 'To check these predictions we need to understand the shape of a pyramid very carefully — let us build one.'	Predict-Before-Observe (PBO): Activating prior knowledge and surfacing misconceptions before instruction. By committing a prediction to paper, learners create a cognitive hook — they will be motivated to compare their estimate against the derived answer at the end of the lesson. Recording on the board makes the class's thinking visible and creates accountability.	Teacher scans written predictions as learners write. Observable indicators of prior understanding: Do learners estimate only the base area (common misconception — treating a pyramid like a flat square)? Do any learners mention the slant sides? Do learners spontaneously connect to Lesson 1 (same-surface-area-different-volume idea)? Teacher notes which pairs conflate surface area with volume — these pairs will be targeted with probing questions during Phase 3.
Observe Phase 📹 **VIDEO:** Volume of a Pyramid - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four. Each group receives: one pre-scored card net of a square-based pyramid (base 8 cm × 8 cm, slant height 10 cm) ready to fold; one pre-scored card net of a triangular pyramid (equilateral triangular base, side 6 cm, slant height 9 cm); a ruler; scissors. STEP 1 (5 min): Before folding, learners measure and label every face of the flat net — they record the base dimensions and the dimensions of each triangular face in their books. STEP 2 (5 min): Learners fold and tape the nets into 3-D pyramid	Distribute materials before the lesson (one kit per group of 4). As groups fold nets say: 'Your net is the iron sheet before the metalsmith bends it. What shape are all the pieces?' (Expected: one square plus four triangles.) After folding, circulate and ask each group: 'Point to the slant height on your model. Now point to where the vertical height would be.' (WAIT TIME 10 seconds per group.) For any group that confuses the two, hold the vertical stick next to the model: 'The vertical height never touches a	Construct-and-Measure (Hands-on Modelling): Physical net construction makes abstract formulas tangible. Learners encounter the slant height as a real, measurable quantity on their own model before any formula is introduced, reducing the common error of substituting vertical height into the surface area formula. Tracing the net into exercise books creates a personal annotated reference.	Circulate and check: (1) Can each learner correctly point to and measure slant height vs. vertical height on their model? (2) Are the triangular face dimensions labelled correctly on the traced net? (3) Does the learner recognise that there are FOUR identical triangular faces on a square-based pyramid? Flag any group that labels the vertical height as the slant height — address immediately with the stick-and-string demonstration before moving to Phase 3.

	models. They hold up the assembled square-based pyramid and the teacher asks: 'Count the faces out loud together.' Class responds: '5 — one square base and four triangular faces.' STEP 3 (5 min): Teacher holds up a pre-made larger card pyramid with a vertical stick (representing the centre pole) and a coloured string laid along the slant face. Say: 'This stick shows the VERTICAL height — from the apex straight down to the centre of the base. This string shows the SLANT HEIGHT — from the apex down the face to the midpoint of a base edge. Are they the same length?' Learners measure both on their own models and record: vertical height and slant height values. STEP 4 (5 min): Learners lay the assembled model flat, unfold it, and trace the net onto their exercise book page, labelling: base, triangular face, slant height (l), base side (b), vertical height (h).	face — it goes straight up inside the pyramid like a centre pole. The slant height is the distance down the outside of the roof.' Cold-call one group to explain the difference to the class in their own words. Record on the board: SLANT HEIGHT (l) ≠ VERTICAL HEIGHT (h) in a box. Connect to phenomenon: 'The metalsmith needs to know the SLANT height, not the vertical height, to cut iron sheet for each triangular face. Why?'		
Explain Phase 📹 **VIDEO:** Volume of a Pyramid - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Deriving the formula (10 min): Each group receives a 'Formula Derivation Strip' — a half-A4 card with three prompts: (1) 'Area of the square base = ?'; (2) 'Area of ONE triangular face = ?'; (3) 'Total surface area = base + all triangular faces = ?'. Groups work silently for 5 minutes to write expressions using symbols b (base side) and l (slant height). Teacher then cold-calls 3 groups to share their expression for one triangular face. Class agrees: Area of one triangular face = ½ × b × l. Teacher writes on the board step-by-step: SA = b^2 + 4 × (½ × b × l) = b^2 + 2bl. Teacher says: 'Notice — slant height l, not vertical height h. This is the distance DOWN THE	During Part A derivation, circulate and look for groups who write 'Area of triangle = b × h' using vertical height — prompt: 'Which height can you MEASURE on the outside of the roof with a tape measure — the inside vertical pole, or the distance along the slant face?' (WAIT TIME 12 seconds.) Do NOT give the answer; redirect to their physical model. During Part B, after revealing SA = 96 m^2, return to the prediction board: 'We predicted [X]. We calculated 96 m^2. What information did those who underestimated leave out?' (Expected: they forgot the triangular faces / used the wrong height.) During Part C, announce:	Guided Inquiry with Structured Derivation (Formula Derivation Strip): Rather than presenting the formula, learners build it from the net they already traced. This reflects NGSS Science and Engineering Practice 2 (Developing and Using Models) and Practice 8 (Obtaining, Evaluating, and Communicating Information). The contextualised worked example (school hall in Eldoret) immediately grounds the abstract formula in the driving phenomenon, and the group practice problems reinforce transfer to novel Kenyan contexts.	Collect Formula Derivation Strips at the end of Part A — scan for the most common errors: (a) using vertical height instead of slant height, (b) forgetting the base area when computing total surface area (or including it when only roof faces are needed), (c) counting only 2 triangular faces instead of 4. Use these errors to frame the debrief of group practice problems. During Part C, observe whether learners read the problem carefully enough to determine whether the base should be included (ceiling vs. open base).

	ROOF FACE — the measurement the metalsmith measures with a tape on the actual roof.' PART B — Worked example: School Hall Roof in Eldoret (8 min): Teacher works through the prediction photograph: base = 6 m, slant height = 5 m (calculated from Pythagoras: $l = \sqrt{h^2 + (b/2)^2} = \sqrt{4^2 + 3^2} = 5$ m — teacher shows this step explicitly on the board). $SA = 6^2 + 2 \times 6 \times 5 = 36 + 60 = 96$ m². Teacher asks: 'Look at your prediction on the board. Who was closest? What did you miss if you underestimated?' PART C — Group practice problems (7 min): Each group solves one of two contextualised problems: GROUP A/C: 'A community hall near Kericho has a square-based pyramid roof with base side 8 m and slant height 6.5 m. Iron sheet costs KSh 850 per m². Find the total cost of iron sheet for the four triangular roof faces only (not the base — the base is the ceiling).' GROUP B/D: 'A triangular pyramid bandas roof near Malindi has an equilateral triangular base of side 5 m and each triangular face has a slant height of 4.2 m. Find the total surface area of all four faces.' Groups write solutions on the board; class reviews together.	'You have 7 minutes. Every group member must write the solution in their own book — not just the fastest writer.' After 7 minutes, cold-call one non-volunteer from each group to present a step. Connect to phenomenon after Part C: 'The metalsmith at the posho mill in Eldoret cuts iron sheet for roofs AND for storage containers. For the roof he needs surface area to know how much sheet to buy. Why does the mill owner care about something DIFFERENT from surface area when it comes to grain storage?' (Anticipated: the mill owner cares how much grain fits INSIDE — that is volume, not surface area.) Record this contrast on the DQB in Phase 4.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Volume of a Pyramid - Overview] Source: CK-12 🔍 Search ARES for similar videos	Each learner independently writes two sticky notes (or index-card slips): STICKY NOTE 1 (GREEN — 'What I now understand'): One sentence connecting today's learning to the posho mill phenomenon. Example responses: 'Surface area of a pyramid tells the metalsmith how much iron sheet to buy for the roof — just like it told us how much sheet was used for the drum and the box.' STICKY	Give learners 3 minutes to write independently (no pair discussion yet — individual accountability). Then allow 2 minutes to walk to the DQB and post notes. Read 3–4 orange notes aloud verbatim without attributing them to individuals — preserve psychological safety. Point to the two most powerful questions and say: 'This question — [reads question about same surface area,	Driving Question Board — Cumulative Evidence Tracking: The DQB functions as a public, evolving record of the class's growing understanding of the phenomenon. By physically adding to it each lesson, learners experience science/mathematics as a progressive sense-making process (NGSS Practice 8). The two-colour sticky note system distinguishes consolidated	Teacher reads all green notes before the next lesson to assess whether learners correctly distinguish surface area (material/covering) from volume (interior capacity). Red flag: any green note that conflates the two (e.g., 'Surface area tells us how much grain the pyramid holds') — these learners need a targeted clarifying conversation at the start of Lesson 3. Teacher photographs

📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	NOTE 2 (ORANGE — 'A new question I have'): One genuine question that today's lesson raised. Example responses: 'If the pyramid roof and the box roof are made from the same area of iron sheet, does the pyramid roof enclose more space inside?' / 'How does the artisan measure slant height on a real roof without climbing up?' / 'What if the base is not square — how does the formula change?' Learners place notes on the class DQB under two columns: WHAT WE ARE FIGURING OUT and QUESTIONS WE STILL HAVE. Teacher reads 3–4 orange notes aloud, circles two that will be explicitly addressed in upcoming lessons (cone surface area; volume), and says: 'These questions are exactly what scientists and engineers ask. Keep them — we will answer them lesson by lesson.'	different enclosed space] — is the heart of what the posho mill owner is trying to understand. We will return to it when we study volume.' Draw an explicit arrow on the DQB from the Lesson 1 entry ('Same sheet metal, different grain capacity') to today's entry ('Pyramid roof surface area = material used, not space inside'). This visual progression shows learners that the DQB is a growing explanation, not a list of isolated facts.	understanding (green) from productive uncertainty (orange), modelling how mathematicians distinguish proven results from open conjectures.	the DQB board to track progression across the 8-lesson sequence.
Model Building Phase 📹 **VIDEO:** [Volume of a Pyramid - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the cumulative class model begun in Lesson 1 — a large annotated diagram on A2 paper (or the chalkboard) showing the posho mill's cylindrical drum and rectangular box, labelled with surface area = same for both. Today learners ADD a third element to the model: they sketch a pyramid and annotate it with: (1) the formula SA = $b^2 + 2bl$; (2) an arrow labelled 'slant height $l \neq$ vertical height h'; (3) a speech bubble from the metalsmith figure: 'I use SA to know how much iron sheet to cut — whether for a roof or a storage container.' Below the model, learners individually write a one-sentence 'So far' claim: 'Surface area measures the material covering the outside of a solid; it does not tell us how much	Display the Lesson 1 cumulative model at the front. Say: 'Our model already shows two container shapes — the drum and the box. Today we discovered a third shape used by the same metalsmith — the pyramid roof. Add the pyramid to your model with its formula and the slant-height annotation.' Allow 5 minutes for individual model revision. Then cold-call one learner to read their 'So far' claim aloud. Ask the class: 'Do you agree? Does surface area tell us how much grain fits inside? Thumbs up / thumbs down.' (Expected: thumbs down — surface area does not tell volume.) Say: 'Exactly. In the next lessons we will measure what IS inside these shapes — the volume. That is what the posho mill owner really	Cumulative Model Revision (Progressive Model Building — NGSS Practice 2): The class model is a living artefact that grows each lesson, making the progression of understanding visible. Adding the pyramid alongside the drum and box reinforces the unit's central contrast: same surface area does not mean same volume. The metalsmith speech bubble humanises the mathematics and anchors it in the Kenyan artisan context of the driving question.	Teacher initials books only where: (a) the pyramid sketch shows both base and triangular faces; (b) the annotation correctly distinguishes slant height from vertical height; (c) the 'So far' claim correctly states that surface area ≠ volume / capacity. Books without the initials are flagged for a 2-minute catch-up conversation at the start of Lesson 3. This creates a low-stakes accountability loop that does not slow the lesson but ensures no learner silently misunderstands the core distinction.

	grain (volume) it can hold.' Teacher circulates and stamps (or initials) books where the claim is correctly written.	needs to understand.' End by pointing to the DQB: 'Our orange questions are our roadmap for the rest of this unit.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners successfully distinguish slant height from vertical height during the net-construction activity, or did the majority still substitute vertical height into the formula during group practice? If so, plan a 5-minute re-teach at the start of Lesson 3 using the physical model and the Pythagorean relationship explicitly. (2) Which groups produced the clearest Formula Derivation Strips? Consider using these as exemplars (anonymised) at the start of the next lesson to model mathematical reasoning. (3) Were the contextualised problems (Kericho hall, Malindi bandas) sufficiently challenging, or did all groups finish with more than 2 minutes to spare? If so, prepare an extension problem involving a frustum (a pyramid with the top cut off — common in Kenyan water tank lids) for fast finishers in Lesson 3. (4) Review the orange DQB sticky notes: How many learners asked about volume vs. surface area? This is the conceptual pivot of the entire 8-lesson sequence — high frequency of this question signals the class is ready for an accelerated transition to volume concepts. (5) Did all learners record the 'So far' claim accurately? The claim 'Surface area measures the material covering the outside — not the space inside' is the foundational conceptual anchor for the posho mill phenomenon resolution.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners constructed card nets of square-based and triangular pyramids, folded them into 3-D models, and measured both the slant height and the vertical height on their models. They observed that the slant height (the distance along the outer face from apex to base-edge midpoint) is always longer than the vertical height (the straight inner height), and that a square-based pyramid has one square face and four identical triangular faces.
What did I learn?	Learners derived the surface area formula for a square-based pyramid: $SA = b^2 + 2bl$ (where b = base side length and l = slant height). They learned that slant height must be used — not vertical height — because it is the dimension that lies on the outer face of the pyramid, representing the material the metalsmith actually cuts. They applied Pythagoras' theorem to calculate slant height from given vertical height and base dimensions, and solved real-life problems involving the cost of iron sheet for pyramid-shaped roofs in Eldoret and Kericho.
How does this explain the phenomenon?	Surface area measures the total area of the material covering the outside of a solid (like the iron sheet a metalsmith cuts to make a pyramid roof or a storage container). It does NOT measure how much space is inside the solid — that is volume. The posho mill owner's mystery cannot be solved by surface area alone: two containers can be made from the same area of iron sheet (same surface area) but hold very different amounts of maize (different volumes). To explain why the cylindrical drum holds more grain than the rectangular box, we must investigate volume — which is what the next lessons will explore.

LESSON 3 (80 minutes): Surface Area of Cones and Frustums: Unrolling the Cone to Find the Metal Sheet	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to derive and apply the curved surface area formula for cones and frustums using nets, connecting to the posho mill phenomenon and the Turkana water-tank context.
Knowledge	Learners know that the net of a cone is a sector of a circle. They know the relationship between the slant height (l), radius (r), and height (h) of a cone via Pythagoras' theorem. They know that a frustum is a cone with its apex removed, and can express its curved surface area as the difference of two conical surfaces: $\pi R l_1 - \pi r l_2$, simplified using similar-triangle ratios. They know that total surface area of a closed cone = $\pi rl + \pi r^2$, and that of a frustum = $\pi(R + r)l + \pi R^2 + \pi r^2$, where l is the slant height of the frustum.
Skills	Learners can: (a) construct the net of a cone from a given sector and identify its dimensions; (b) derive the curved surface area formula CSA = πrl by relating arc length of the sector to the circumference of the circular base; (c) apply Pythagoras' theorem to find slant height; (d) calculate the curved and total surface area of a cone and frustum; (e) determine the amount of metal sheet needed for a conical or frustum-shaped tank top, linking back to the posho mill phenomenon.
Attitudes	Learners appreciate that mathematical derivation from first principles (unrolling a cone into a flat net) gives artisans and engineers a reliable formula — reinforcing the value of integrity in craftsmanship and the importance of precision in construction. Learners show responsibility by connecting their calculations to real resource planning for Kenyan artisans.
Key Inquiry Question	How do we determine the surface area of solids? (KICD Key Inquiry Question 1) | How are surface area and volume of solids applied in real-life situations? (KICD Key Inquiry Question 2)
Purpose in Storyline	In Lessons 1–2 learners established that equal surface area does NOT mean equal volume for cylinders and cuboids. The posho mill owner's puzzle remains open. Lesson 3 widens the investigation: the metalsmith in Eldoret also makes conical funnel tops for grain-pouring spouts and frustum-shaped water-tank roofs for a community project in Turkana. Before the artisan can cut the iron sheet, they need to know exactly how much sheet metal is required. Learners derive the formula by unrolling the cone — the same logical move a metalsmith makes when marking out a flat sheet before bending it. This keeps the driving question alive: shape determines both how much material you need AND how much a container holds.
Safety Notes	No hazardous materials are used. Scissors and compasses have sharp points — remind learners to handle them carefully and to pass scissors handle-first. Ensure adequate desk space when learners are unrolling and cutting paper nets to avoid accidental cuts.

B. LESSON OVERVIEW
Lesson 3 of 8 in the evidence-gathering sequence for Sub-Strand 2.7. Learners begin by predicting how much paper is needed to wrap a conical ice-cream cone, connecting to the posho mill artisan phenomenon. They then observe and physically construct the net of a cone (a sector of a circle) to derive the curved surface area formula CSA = πrl from first principles — matching the arc length of the sector to the base circumference. They extend the derivation to the frustum by treating it as a large cone minus a small cone, and calculate the metal sheet needed for a conical grain-funnel and a frustum-shaped Turkana water-tank roof. The lesson deepens the storyline: the metalsmith must know surface area precisely before cutting iron sheet, while the driving question reminds learners that surface area and volume remain distinct, shape-dependent properties.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Writing proportional equations from tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Volume of the Cone (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a flat circular piece of paper (pre-cut, radius 12 cm) and a small conical party hat or an empty ice-cream cone. The teacher poses: 'This cone was made from flat sheet — just like the artisan in Eldoret cuts iron sheet. Without measuring anything yet, sketch on your whiteboard or notebook: what shape do you think the flat piece of paper was before it was rolled into this cone? Label any lengths you think matter.' Learners work individually for 3 minutes, then share predictions with a shoulder partner. The teacher takes a quick show-of-hands poll: 'How many drew a rectangle? A triangle? A sector of a circle? A full circle?' Predictions are recorded on the class Driving Question Board under the column 'We Think…'	Distribute one cone and one pre-cut paper circle per pair before the lesson starts to save time. After posing the prediction prompt, say: 'Take 3 minutes — silent thinking first, then discuss with your partner.' Set a visible timer. After 3 minutes say: 'Hands up if you drew a full circle.' Count aloud ('I see 4, 5, 6…'). Repeat for rectangle, triangle, sector. Cold-call 3 learners from different predictions to justify briefly: 'Amina, you drew a sector — tell us in one sentence why.' Provide 10 seconds wait time after each cold-call before accepting the answer. Do NOT confirm or correct at this stage — record all predictions neutrally on the board. Anchor to phenomenon: 'The metalsmith in Eldoret faces exactly this question every time he makes a conical funnel top. Let us find out who is right.'	Predict–Observe–Explain (POE) activation: By making a concrete prediction before instruction, learners activate prior knowledge about circles and sectors. Recording predictions publicly creates cognitive dissonance that motivates the derivation activity. This strategy ensures learners are not passive recipients of the formula but are invested in resolving their own prediction.	Observable indicator: Learners produce a labelled sketch within 3 minutes. Listen for use of vocabulary — 'radius', 'arc', 'sector' — as evidence of prior knowledge from circle work (Sub-Strand 2.6). If fewer than 30% sketch a sector, pause and ask: 'Think about what happens when you unroll a cone — what 2-D shape do you get?' This signals whether a brief vocabulary bridge on sectors is needed before Phase 2.
Observe Phase 📹 **VIDEO:** Writing proportional equations from tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Volume of the Cone (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Activity 2A — Unrolling the Cone (15 min): In pairs, learners take their pre-cut paper circle (radius L = 12 cm), cut out a sector of angle θ (sectors are pre-marked at 240° to give a realistic cone), and roll it into a cone. They measure: the slant height l (radius of the sector = 12 cm), the radius r of the circular base, and the arc length of the sector. They record values in a structured table: | Sector radius (l) | Sector angle θ | Arc length of sector | Circumference of base circle (2πr) | Ratio arc/circumference |. Learners compare the arc length of the sector with the circumference of the base circle and record what	Before Activity 2A, demonstrate the cutting and rolling with your own copy — hold up the sector and say: 'Notice I am NOT measuring the cone yet. I am just observing what shape I get and what measurements seem related.' Circulate during the 15 minutes; at the 7-minute mark do a mid-activity cold-call: choose 2 pairs and ask 'What did you notice about the arc length and the base circumference?' Wait 10–12 seconds before pressing for an answer. If a pair says 'they are equal' or 'very close', ask 'Why would that be? Think about how the sector wraps around.' After the video, say: 'On your sticky note,	Concrete–Representational–Abstract (CRA) progression: Learners begin with a physical object (concrete cone), move to a flat net with measurements (representational table), and will move to an algebraic formula (abstract) in Phase 3. The video provides a real-world anchor that keeps the derivation purposeful. The sticky-note bridge between video and paper activity reinforces transfer.	Observable indicator: The table column 'Ratio arc/circumference' should show values close to 1.0 for all pairs (since the arc of the sector IS the circumference of the base). Circulate and check tables — if a pair's ratio is not close to 1, check whether they measured the arc correctly (use a string along the curved edge). Listen for the phrase 'they are the same' or equivalent — this is the key conceptual observation that drives the derivation. If fewer than half the pairs reach this observation by the 12-minute mark, pause the class and orchestrate a whole-class share.

	they notice. Activity 2B — Video clip (5 min): Learners watch a 2-minute ARES platform video showing a Turkana community metalsmith marking out and rolling a conical roof section for a rainwater-harvesting tank, narrated in Swahili with English subtitles. The clip shows the metalsmith using a rope compass to draw a large sector on a sheet of iron. Learners note on a sticky note: one thing they observed the metalsmith doing that connects to their paper activity.	write one connection between the metalsmith's action and your paper cone. You have 90 seconds.' Cold-call 3 learners to share. Anchor to phenomenon: 'This is exactly what the Eldoret artisan does — before bending the iron sheet, he marks out a sector. Our job now is to turn what we observed into a formula he can use every time.'		
Explain Phase 📹 **VIDEO:** [Writing proportional equations from tables] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Volume of the Cone (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	Derivation (whole class, guided): The teacher leads a step-by-step derivation on the board, with learners filling in a structured derivation scaffold in their notebooks. Steps: (1) The net of a cone is a sector of radius l (slant height) and angle θ. Area of sector $= (\theta/360°) \times \pi l^2$. (2) The arc length of the sector $= (\theta/360°) \times 2\pi l$. (3) From observation: arc length = circumference of base $= 2\pi r$, therefore $(\theta/360°) = r/l$. (4) Substitute into area of sector: $CSA = (r/l) \times \pi l^2 = \pi rl$. ✓ Formula confirmed from first principles. (5) Total surface area of a closed cone: $TSA = \pi rl + \pi r^2$. Learners label a diagram of the cone with r, l, h and use Pythagoras to write $l = \sqrt{(r^2 + h^2)}$. Application Task A — Conical Funnel (pair work, 8 min): 'The Eldoret metalsmith makes a conical grain-funnel with base radius 21 cm and vertical height 28 cm. Calculate: (i) the slant height, (ii) the curved surface area of sheet metal needed.' [Answer: $l = \sqrt{(21^2 + 28^2)} = \sqrt{(441 + 784)} = \sqrt{1225} = 35$ cm; $CSA = \pi \times 21 \times 35 = 735\pi \approx 2309$ cm²] Application Task B — Frustum Extension	During the derivation, pause after step (3) and cold-call 2 learners: 'Kariuki, what does $(\theta/360°)$ represent geometrically?' Wait 12 seconds. After step (4) say: 'Turn to your partner — convince them in 30 seconds why this substitution works.' Then cold-call one pair to share their explanation. Write the final formula in a box: $CSA = \pi rl$. Say: 'This is the formula the metalsmith uses. Notice it came directly from what you observed — the arc length equals the base circumference.' For Application Task A, say: 'Work in pairs. You have 8 minutes. Show all working — the artisan needs to see every step to avoid wasting expensive iron sheet.' At 5 minutes, circulate and check slant height calculations — common error is forgetting Pythagoras. For the frustum, use the questioning sequence: 'If I stack the small cone back on the frustum, what do I get? (A full cone.) So what is the CSA of the frustum in terms of two conical surfaces?' Wait 15 seconds, cold-call 3 learners before guiding to the answer. Anchor to phenomenon: 'Now the Eldoret	Derivation-first, formula-second (Inquiry-based mathematical modelling): Learners derive the formula rather than receive it, directly connecting the algebraic result to their physical observation in Phase 2. This embeds SEP-2 (Developing and Using Models) and SEP-5 (Using Mathematics). The two-tier application (cone → frustum) uses scaffolded abstraction: the frustum formula is presented as a logical extension, not a new memorised rule, building structural understanding.	Observable indicator: In Task A, monitor whether learners correctly identify l as the slant height (not the vertical height h) in the formula $CSA = \pi rl$. A common error is substituting h = 28 directly. If more than 3 pairs make this error, pause and re-draw the cone cross-section, asking: 'Which measurement does the iron sheet follow — the vertical height or the slanted side?' For Task B, check that the independent frustum answer is $\pi(45+15)\times 40 = 2400\pi \approx 7540$ cm². Exit-check: before moving to Phase 4, ask all learners to hold up their notebook showing the boxed formula and one completed calculation — a quick visual scan identifies who needs support.

	(pairs, 10 min): Teacher draws a frustum on the board. 'A frustum is a cone with the top cut off. The Turkana water-tank roof is a frustum with large radius R = 60 cm, small radius r = 20 cm, and slant height l = 50 cm. How much iron sheet is needed for the curved surface only?' Teacher guides derivation: CSA of frustum = $\pi(R + r)l = \pi(60 + 20) \times 50 = 4000\pi \approx 12\,566$ cm². Learners work through a second frustum problem independently: R = 45 cm, r = 15 cm, l = 40 cm.	artisan and the Turkana community can both calculate exactly how much metal sheet they need — no waste, no shortage.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Writing proportional equations from tables] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Volume of the Cone (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board. In groups of 4, they discuss and write responses on sticky notes for two columns: (1) 'What we have NOW learned that helps answer the DQ' — e.g., 'Surface area of a cone depends on both r and l — shape AND size matter, not just the amount of material.' (2) 'New questions this raises' — e.g., 'Does a cone hold more or less grain than a cylinder made from the same sheet area? We haven't compared volumes yet.' Groups post their sticky notes. The teacher facilitates a 4-minute gallery walk where learners read other groups' contributions and place a small star sticker on any new question they find most interesting for future lessons. The teacher highlights two sticky notes that directly address the posho mill phenomenon: 'Same metal area, different shape = different amount of grain' and 'We still need to find volumes of cones and frustums to fully solve the puzzle.'	Say: 'You have 5 minutes in your groups. Look at the DQ: the posho mill owner wants to know why the cylinder holds more grain. Today's lesson gives us part of the answer — what part? Write it on a sticky note.' Circulate, listening for groups who conflate surface area with volume — gently redirect: 'Be precise — did we calculate how much grain fits inside, or how much metal sheet is needed?' After posting, facilitate the gallery walk with the instruction: 'Read silently for 2 minutes, then star the new question you most want answered in a future lesson.' Cold-call 2 learners to read their starred question aloud and explain why it matters. Close by pointing to the DQB: 'Notice our board is growing — we are solving one piece of the puzzle each lesson. Today: surface area of cones and frustums. Still to come: their volumes.' Record a summary statement on the board: 'Lesson 3 evidence: CSA = πrl (cone); CSA = $\pi(R+r)l$ (frustum). Surface area ≠ volume — the puzzle is not yet fully solved.'	Driving Question Board as a living epistemic artefact: The DQB externalises the class's growing understanding and makes the gap between current knowledge and the full explanation of the phenomenon visible and explicit. Posting 'new questions' models the scientific practice of recognising the limits of current evidence (SEP-6: Constructing Explanations), motivating the remaining lessons in the sequence.	Observable indicator: Each group's 'What we learned' sticky note should distinguish surface area from volume and reference the cone/frustum formula with correct notation. If a group writes only 'we learned CSA = πrl' without connecting to the phenomenon, prompt: 'How does this help the posho mill owner or the Turkana community?' The quality of 'new questions' reveals depth of thinking — surface-level questions ('what is the formula for a sphere?') versus structural questions ('does a cone with the same surface area as a cylinder always hold less volume?') indicate different levels of conceptual engagement. Note group names for targeted support in Lesson 4.

Model Building Phase 📹 **VIDEO:** [Writing proportional equations from tables] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Volume of the Cone (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner updates their personal 'Solids Surface Area Model' — a cumulative annotated diagram started in Lesson 1. Today they add: (a) a labelled cross-section of a cone showing r, h, l with the Pythagoras relationship $l = \sqrt{r^2 + h^2}$; (b) the net of the cone (sector) with annotations showing arc length = $2\pi r$ and sector radius = l; (c) the formula CSA = πrl and TSA = $\pi rl + \pi r^2$ in a colour-coded box; (d) a labelled frustum diagram with R, r, l and the formula CSA = $\pi(R + r)l$; (e) a one-sentence connection to the phenomenon: 'The Eldoret metalsmith uses CSA = πrl to calculate how much iron sheet to cut before rolling the cone — shape determines the sheet needed, just as shape determines the volume stored.' Learners compare their updated model with a partner and identify one thing their partner included that they missed. The teacher collects 6 models (2 high, 2 middle, 2 developing) for written feedback to be returned at the start of Lesson 4.	Say: 'Open your model from Lesson 1. You are the artisan's technical advisor — update your model so the metalsmith can use it as a reference card.' Allow 8 minutes of individual work. Play quiet background ambient sound (optional, if available on ARES platform). At 6 minutes, say: 'Share your model with your partner. Find one thing they drew or wrote that you did not. Add it to your own model.' After 2 minutes, say: 'In one sentence at the bottom of your model, write how today's lesson connects to the posho mill owner's puzzle.' Cold-call 3 learners to read their sentence aloud — listen for the distinction between surface area and volume. Collect 6 models as described. Close the lesson: 'Today we proved that CSA = πrl — not from a textbook, but from unrolling a cone, just like a metalsmith unrolls iron sheet. In Lesson 4, we will find out what these shapes can HOLD — their volume. That will finally start to solve the mill owner's puzzle.'	Cumulative model revision (progressive modelling, aligned with NGSS SEP-2): The personal annotated diagram grows across all 8 lessons, making the architecture of understanding visible to both learner and teacher. Adding a phenomenon-connection sentence each lesson ensures the model is not a passive note but an active explanatory tool. Partner comparison activates metacognition — learners notice gaps in their own representation.	Observable indicator: The model should contain all five elements listed in the learner experience. Prioritise checking: (i) Is l correctly identified as slant height, not vertical height? (ii) Is the net drawn as a sector (not a rectangle or triangle)? (iii) Does the frustum diagram correctly label both R and r? (iv) Does the phenomenon-connection sentence use the word 'surface area' correctly (not conflated with volume)? Use a 3-column feedback stamp on collected models: 'Strong ✓', 'Needs label correction ✓?', 'See me ✗'. Return with written comment at the start of Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners successfully observe that the arc length of the sector equals the circumference of the cone's base during Activity 2A? If fewer than 60% of pairs reached this observation independently, consider providing a more structured measurement guide in future iterations. (2) Were learners able to distinguish between slant height (l) and vertical height (h) when applying CSA = πrl? This is the most common error in this lesson — if it persisted beyond the first application task, plan a focused 5-minute warm-up at the start of Lesson 4 using a labelled diagram. (3) Did the frustum derivation feel rushed? If the 10-minute allocation was insufficient, consider splitting the frustum into the first 10 minutes of Lesson 4, treating it as a predict–derive warm-up linked to the Turkana water-tank context. (4) Were learners' DQB sticky notes distinguishing surface area from volume? The success of Lessons 4–5 (volume of cones and frustums) depends on learners holding this distinction clearly. (5) Review the 6 collected models before Lesson 4 — note patterns in misconceptions (e.g., using diameter instead of radius, omitting the base circle in TSA) and address these explicitly in the Lesson 4 opening warm-up.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners physically unrolled a paper cone into a sector and measured its arc length and the base circumference, observing that the two lengths are equal. They also watched a video of a Turkana metalsmith marking out a conical roof sector on a flat iron sheet.
What did I learn?	The net of a cone is a sector of a circle with radius equal to the cone's slant height l. Because the arc of the sector equals the base circumference ($2\pi r$), the curved surface area of a cone is $CSA = \pi rl$. The slant height is found using Pythagoras: $l = \sqrt{(r^2 + h^2)}$. A frustum's curved surface area is $CSA = \pi(R + r)l$, derived by subtracting two conical surfaces. These formulas allow a metalsmith to calculate exactly how much iron sheet to cut before shaping a cone or frustum.
How does this explain the phenomenon?	The posho mill artisan in Eldoret uses $CSA = \pi rl$ to determine the sheet metal needed for a conical funnel top, and $\pi(R + r)l$ for a frustum-shaped water-tank roof. This confirms that the same total area of metal sheet can be formed into very different shapes — and that the shape of a container, not just the sheet area used, determines how much grain it can hold. The driving question remains partially open: we have now compared surface areas of cones and frustums, but we have not yet compared their volumes to those of the cylinder and cuboid from Lessons 1–2. Volume calculations in Lessons 4–5 will complete the explanation of why the cylindrical drum holds more maize than the rectangular box.

LESSON 4 (80 minutes): Surface Area of Spheres and Composite Solids	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners extend their understanding of surface area to spheres and composite solids, discovering that shape complexity — not just material used — determines how much iron sheet is needed to enclose a given space, directly advancing the posho mill driving question.
Knowledge	By the end of this lesson, learners should be able to: (a) derive and apply the formula SA = $4\pi r^2$ for the surface area of a sphere; (b) identify the component surfaces of composite solids (cylinder + cone; hemisphere + cylinder); (c) calculate the total surface area of composite solids by decomposing them into known shapes.
Skills	Measuring, decomposing composite solids into component shapes, applying formulae, communicating mathematical reasoning, collaborative problem solving.
Attitudes	Appreciate that mathematical formulae for curved surfaces reflect real engineering decisions; show curiosity about why spherical shapes appear in packaging, sports equipment, and storage design across Kenya.
Key Inquiry Question	How do we determine the surface area of solids? How are surface area and volume of solids applied in real-life situations?
Purpose in Storyline	In Lesson 3 learners calculated the surface area of cones and frustums. Now they tackle spheres — the ultimate curved solid — and composite shapes that combine solids. This unlocks the question: if a Nairobi packaging factory or a Kisumu sports-equipment maker wraps a spherical or dome-topped container, how much material do they need? Learners update the DQB with the new insight that surface area formulae depend on the geometry of every face, including curved ones, and that composite solids require decomposition before calculation.
Safety Notes	Scissors and craft knives used in the paper-wrapping activity must be handled with care. Learners should keep blades pointed away from themselves and others. Teacher supervises all cutting. No safety concerns with footballs or cylindrical models.

B. LESSON OVERVIEW

Learners begin by predicting how many flat circular discs of paper (radius r) would be needed to cover a football (radius r). They then conduct a hands-on wrapping investigation — cutting great-circle discs and tearing them into strips to cover a ball — discovering empirically that exactly four discs cover the sphere, verifying SA = $4\pi r^2$. The class applies this to a Nairobi packaging factory context (spherical tins, dome-topped containers) and a Kenyan football manufacturing scenario. In the Explain phase, learners decompose composite solids (a grain silo modelled as cylinder + cone, and a gas cylinder modelled as cylinder + two hemispheres) into components, calculate each surface area, and combine. The DQB is updated with the new understanding that curved-surface solids require specialised formulae and that composite shapes must be decomposed. A model revision closes the lesson.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Writing]	Each learner receives a printed circle (radius r, where r matches the football used in the next	Distribute one pre-cut circle and one football (or spherical plastic ball) to each group of four. Say:	Prediction with Justification — learners commit a specific numerical estimate with reasoning	Teacher circulates and scans written predictions. Look for: (1) Does the learner give a numerical

proportional equations from tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	phase) and a football or spherical ball. Learners individually write a prediction in their exercise books: 'How many of these flat circular discs do you think would be needed to completely cover the surface of this ball?' They also sketch what they think the 'unrolled' surface of a sphere might look like, and write one sentence connecting their prediction to the posho mill phenomenon: 'How might the shape of a spherical container affect how much iron sheet a metalsmith in Eldoret would need?'	'Before we do anything, I want your best mathematical guess — written down, committed to paper. Look at this circle and look at this ball. How many of these circles would tile the ball perfectly?' Allow 10 seconds WAIT TIME in silence. Then say: 'Write your number and one sentence of reasoning. You have 3 minutes — go.' After 3 minutes, cold-call 5 learners using name cards: ask each for their number AND their reasoning. Record all predicted numbers on the board (expected range: 2–8). Do NOT confirm or deny. Say: 'We are about to find out who is closest. Hold your prediction — we will return to it.' Connect to phenomenon: 'Remember our posho mill owner in Eldoret — he is puzzled about metal sheet and capacity. Today we ask: for a spherical container, how much sheet metal is actually needed?'	before instruction, activating prior geometric intuition about curved surfaces and creating cognitive dissonance that motivates the investigation.	estimate (not just 'a lot')? (2) Is the reasoning geometric (mentions radius, circumference, area) or purely intuitive? Learners who write '4' with geometric reasoning are ready for extension. Learners who write '2' (confusing diameter-circle with surface) need targeted support in Phase 2.
Observe Phase 📹 **VIDEO:** Writing proportional equations from tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four conduct the 'Four Circles' wrapping investigation. Each group receives: one spherical ball (football or plastic ball) of radius r, four pre-cut paper circles each of radius r, scissors, tape, and a marker. Step 1 — Learners measure the radius of the ball using a tape measure and record it. Step 2 — They tear or cut the first circle into small strips/pieces and press them onto the ball with tape, covering as much surface as possible. Step 3 — They continue with the second, third, and fourth circles, noting when the ball is completely covered. Step 4 — They record in a table: 'Number of circles used | Fraction of ball covered'. Step 5 — Learners watch a 4-minute teacher-narrated video clip (or	Before the activity, pre-cut circles to match each group's ball radius (measure balls the day before). Say: 'Your task is to cover the entire ball using only these circles — cut or tear them as needed. Count how many full circles you use.' Circulate every 3 minutes. At the 8-minute mark, cold-call 3 groups: 'Group near the window — how many circles have you used so far, and how much of the ball is covered?' After all groups finish (approx. 15 minutes), facilitate a whole-class share. Ask: 'Who predicted exactly 4? Raise your hand.' Then say: 'Let us write what we observed as a mathematical statement.' Write on board: SA = $4\pi r^2$. Ask: 'Why does this make sense given what you just did physically?' Allow 15 seconds	Hands-On Empirical Investigation followed by Formula Derivation — learners physically discover the $4\pi r^2$ relationship before it is stated, so the formula is anchored to a concrete, memorable experience. This is an NGSS Science and Engineering Practice: Planning and Carrying Out Investigations.	Observe group data tables. Correct outcome: exactly 4 circles used to cover the sphere. If a group records 3 or 5, check whether their circles were the correct radius — common error is using circumference instead of radius. Ask: 'Show me how you measured r.' Learners should be able to state SA = $4\pi r^2$ and substitute their measured value correctly within 2 minutes of the share-out. Exit check: teacher points to a sphere drawn on the board labelled r = 7 cm and asks three randomly selected learners to call out the surface area. Correct answer: $4\pi(7^2)$ = 616 cm².

	teacher demonstration) showing the same experiment with a larger ball, confirming the result. Step 6 — Groups record the observation: 'It took exactly ___ circles of radius r to cover the sphere of radius r.' They then write the area formula derivation: Area of one circle = πr^2, so SA of sphere = 4 × πr^2 = $4\pi r^2$. Step 7 — Using their measured radius, groups calculate the actual surface area of their football and compare with a Kenyan standard football ($r \approx 11$ cm): SA = $4\pi(11)^2 \approx 1{,}520$ cm². Context bridge: 'A factory in Nairobi's Industrial Area that manufactures leather footballs needs to cut exactly this much leather per ball. How does knowing SA = $4\pi r^2$ help them minimise leather waste?'	WAIT TIME. Cold-call 2 learners. Then connect: 'This is the same formula a metalsmith in Nairobi or Eldoret uses when fabricating a spherical gas cylinder or a dome-shaped water tank. The mathematics you just discovered with paper and a ball is engineering mathematics.' Pause and ask: 'How does this connect to our posho mill owner's problem about iron sheet and capacity?'		
Explain Phase 📹 **VIDEO:** [Writing proportional equations from tables] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners work through two structured problems connecting to Kenyan real-life composite solid contexts, applying decomposition as the core strategy. PROBLEM 1 — Grain Silo (Cylinder + Cone): A grain cooperative in Nakuru builds a metal silo consisting of a closed cylinder (radius 1.4 m, height 3 m) topped by a conical roof (same radius, slant height 2 m). Calculate the total external surface area of metal sheet needed (excluding the base, which sits on a concrete platform). Learners decompose: (a) Curved surface of cylinder = $2\pi rh$; (b) Circular top of cylinder — EXCLUDED because the cone sits on it; (c) Curved surface of cone = πrl; (d) Base of cylinder = πr^2 — excluded (on ground). Total = $2\pi(1.4)(3) + \pi(1.4)(2)$. Learners calculate and confirm ≈ 35.2 m². PROBLEM 2 — Dome-Topped Water Tank (Hemisphere + Cylinder): A water tank	Write on the board: 'DECOMPOSE → IDENTIFY FACES → SUBTRACT HIDDEN FACES → CALCULATE → COMBINE.' Say: 'Every composite solid problem follows this five-step process. Write it in your books — this is your algorithm.' For Problem 1, display a labelled diagram of the Nakuru silo. Ask: 'Before you calculate anything — which faces are exposed to the outside? Which faces are hidden?' Allow 15 seconds WAIT TIME. Cold-call 4 learners to identify one face each. Confirm the list on the board. Say: 'Now calculate each face area independently, then add. Go.' Circulate and watch for the most common error: including the top circle of the cylinder (hidden under cone). After 8 minutes, pause the class: 'Hands up — who included the top circle of the cylinder? Let us talk about why that is a hidden face.' For Problem 2, repeat the	Decomposition Algorithm with Annotated Diagrams — learners use a named five-step process (Decompose → Identify → Subtract → Calculate → Combine) anchored to labelled diagrams. This makes the abstract process of handling composite solids visible and transferable. NGSS Practice: Using Mathematics and Computational Thinking.	Collect one group's written solution to Problem 1 mid-way through the phase. Check: (1) Are all five decomposition steps present? (2) Is the hidden face correctly excluded? (3) Is arithmetic correct to 2 decimal places? For Problem 2, ask learners to self-mark using an answer displayed on the board. Teacher counts hands for 'fully correct', 'minor arithmetic error', 'wrong face included' — adjust pacing accordingly. Any learner who correctly completes the extension composite solid earns a 'Design Challenge' stamp in their workbook.

	manufacturer in Mombasa makes tanks with a cylindrical body (radius 0.7 m, height 1.2 m) and a hemispherical dome on top (same radius). The base is closed. Find the total surface area of metal sheet needed. Decompose: (a) Base circle = πr^2; (b) Curved cylinder = $2\pi rh$; (c) Hemisphere curved surface = $2\pi r^2$. Note: the flat circle where the hemisphere meets the cylinder is INTERNAL — not included. Total = $\pi(0.7)^2 + 2\pi(0.7)(1.2) + 2\pi(0.7)^2 \approx 8.48$ m². EXTENSION: Learners are asked to sketch and label a composite solid of their own design (e.g., a gas cylinder = cylinder + two hemispheres) and write the surface area formula before calculating.	Decompose step as a whole class before learners calculate independently. Connect to phenomenon after both problems: 'Our posho mill owner in Eldoret wants to build the most efficient container. Now that we can calculate surface area for spheres, cylinders capped with cones, and dome-topped tanks — what does this tell us about how a metalsmith should choose a shape?' Cold-call 3 learners for responses.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Writing proportional equations from tables] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two sticky notes (or index cards): CARD 1 (GREEN — 'What I now understand'): 'I can now calculate the surface area of a sphere using SA = $4\pi r^2$ and decompose composite solids to find their total external surface area.' CARD 2 (YELLOW — 'New question this raises'): A genuine question prompted by today's lesson — e.g., 'If a spherical container has the same surface area as a cylindrical one, does it hold more or less grain?' or 'Why don't posho mills use spherical drums if spheres enclose more volume for the same surface area?' Learners post cards on the class DQB under the columns 'Shape', 'Surface Area', and 'Volume'. The teacher facilitates a 5-minute gallery walk where learners read three peers' yellow question cards and place a tick on any question they also have. The teacher identifies the	Say: 'The DQB is our living record of understanding. In Lesson 1 we asked why the drum holds more grain. In Lessons 2 and 3 we learned to calculate surface area for prisms, cylinders, cones, and frustums. Today we added spheres and composite solids. What has changed in our understanding — and what new mysteries have appeared?' Distribute sticky notes. Allow 4 minutes for writing in silence. Then say: 'Post your green card under the column that matches today's shape. Post your yellow card in the Questions column.' Facilitate the gallery walk with the instruction: 'Read three cards that are NOT yours. Tick any question you share.' After the gallery walk, read the top two ticked questions aloud. Say: 'These are exactly the questions that will drive Lessons 5 and 6, where we move from surface area into volume — and	Driving Question Board Update with Gallery Walk — publicly tracking growing understanding against the anchoring phenomenon makes learning cumulative and visible. Learners generating their own follow-up questions engages NGSS Practice: Asking Questions and Defining Problems.	Teacher reads all green cards after class. Success indicator: green cards include the formula SA = $4\pi r^2$ AND the word 'decompose' or 'composite.' Yellow cards are assessed for quality of mathematical thinking — a high-quality question references both surface area and volume, or makes a comparison between shapes. Cards that only restate today's content ('What is the formula for a sphere?') indicate surface-level processing; these learners are flagged for targeted questioning in Lesson 5.

	top 2 most-ticked questions and writes them prominently on the DQB as 'Our Next Big Questions.'	then compare them. Hold these questions.' Point to the driving question on the wall: 'Why does the same metal sheet make containers that hold different amounts of grain? Are we closer to answering it?' Cold-call 3 learners.		
Model Building Phase 📹 **VIDEO:** [Writing proportional equations from tables](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Surface Area and Nets (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	Learners return to their cumulative 'Container Design Model' — a labelled diagram they have been building since Lesson 1. Today they add a new panel to their model: Panel 4 — Spheres and Composite Solids. In this panel they: (a) Draw a sphere with radius r labelled and write SA = $4\pi r^2$, with a note: 'Discovered by wrapping 4 circles of radius r around a ball'; (b) Draw the Nakuru grain silo (cylinder + cone) with all faces labelled and the formula for total SA written out in decomposed form; (c) Draw the Mombasa water tank (cylinder + hemisphere) with the same treatment; (d) Write one sentence connecting today's learning to the driving question: 'A metalsmith in Eldoret who knows these formulae can calculate exactly how much iron sheet any container shape requires — and can compare shapes to find the most efficient design.' Groups then share their updated models with one neighbouring group, and each group identifies one improvement to make to their model before the next lesson. Teacher collects three models (rotating through groups) for written feedback.	Say: 'Open your Container Design Model. Today we add Panel 4. Your model must show a non-mathematician — like the posho mill owner in Eldoret — exactly how to calculate the surface area of a sphere and a composite solid. Label everything. Write the formula. Show the decomposition.' Allow 10 minutes for model building. Circulate and prompt with: 'Have you shown which faces are hidden in your composite solid?' and 'Can someone reading your model for the first time understand why we use $4\pi r^2$?' At the 8-minute mark, say: 'Exchange your model with the group to your right. Give them one specific piece of feedback: one thing that is clear, and one thing that needs a better label or explanation.' After exchange, say: 'Make that one improvement now.' Collect three models. Close by pointing to the driving question and asking: 'Our posho mill owner now knows how to calculate surface area for a sphere and a composite solid. What is the ONE thing he still does not know that stops him from choosing the best container shape?' Allow 15 seconds WAIT TIME. Cold-call 4 learners. Expected answer: volume — setting up Lesson 5.	Cumulative Model Revision with Peer Feedback — updating a running model across lessons makes conceptual growth visible and forces learners to integrate new knowledge with prior learning. Peer feedback develops communication of mathematical reasoning. NGSS Practice: Developing and Using Models.	Collected models are assessed against three criteria: (1) Formula SA = $4\pi r^2$ correctly stated and connected to the wrapping investigation; (2) Composite solid diagram shows decomposed faces with hidden faces explicitly excluded; (3) A written sentence correctly connects surface area knowledge to the driving question phenomenon. Teacher provides written feedback on collected models before Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did the wrapping investigation produce the empirical result of 4 circles for all groups? If any group got 3 or 5, was this due to measurement error (wrong radius for circles) or procedural error (gaps in coverage)? How will you address this in Lesson 5's opening? (2) In the Explain phase, how many learners correctly excluded the hidden face (top of cylinder in the silo problem) on the first attempt? If more than one-third of learners included hidden faces, consider adding a 'hidden face checklist' to the decomposition algorithm in Lesson 5. (3) Were the DQB yellow cards mathematically rich — did they ask about the relationship between surface area and volume? If most cards only asked about formulae, the class may need a more explicit bridge to the driving question at the start of Lesson 5. (4) Did any learners spontaneously ask why spheres are not used as grain storage containers? This is a high-order question that anticipates the isoperimetric inequality — note these learners for enrichment. (5) Did the Kenyan contexts (Nakuru silo, Mombasa water tank, Nairobi football factory) feel authentic and engaging to learners, or did they seem forced? Adjust naming and scenarios for Lesson 5 based on learner response.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners wrapped a football with paper circles of matching radius and observed that exactly four circles were needed to cover the sphere's surface. They also observed labelled diagrams and worked through calculations for composite solids: a Nakuru grain silo (cylinder + cone) and a Mombasa water tank (cylinder + hemisphere), identifying which faces are exposed and which are hidden.
What did I learn?	The surface area of a sphere is $SA = 4\pi r^2$ — equivalent to four flat circles of the same radius. To find the surface area of a composite solid, learners decompose it into known component shapes, identify all external (exposed) faces, exclude hidden internal faces, calculate each component area using the appropriate formula, and sum the results.
How does this explain the phenomenon?	Learners explained that a metalsmith in Eldoret fabricating a spherical or dome-topped container can use $SA = 4\pi r^2$ and the decomposition algorithm to calculate precisely how much iron sheet is required. They connected this to the driving question by noting that different container shapes — even when built from the same area of iron sheet — enclose different amounts of space, and that knowing surface area formulae for all solid types is a prerequisite for comparing container efficiency. The DQB was updated with new questions about the relationship between surface area and volume, setting up Lessons 5 and 6.

LESSON 5 (80 minutes): Volume of Prisms and Pyramids: How Much Grain Can the Box Really Hold?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners derive and apply the volume formulas for prisms (V = base area × height) and pyramids (V = ⅓ × base area × height) by comparing sand/water-filled physical models, then compute the volume of the posho mill owner's rectangular metal box from Lesson 1 and compare it with the cylinder (to be fully explored in Lesson 6), deepening understanding of why equal surface area does not guarantee equal volume.
Knowledge	By the end of this lesson, learners should be able to: (a) state and derive the formula for the volume of a prism as V = base area × height; (b) state and derive the formula for the volume of a pyramid as V = ⅓ × base area × height; (c) calculate the volume of rectangular prisms (cuboids), triangular prisms, and square-based pyramids; (d) explain why two containers made from the same area of metal sheet can have different volumes.
Skills	By the end of this lesson, learners should be able to: (a) measure and record dimensions of prism and pyramid models accurately; (b) fill models with sand or water and use displacement/direct measurement to compare volumes experimentally; (c) apply volume formulas to solve real-life grain-storage problems set in Kenyan contexts; (d) use prior knowledge of area of polygons (rectangles, triangles) to compute base areas for volume calculations.
Attitudes	Learners should: (a) appreciate that mathematical reasoning — not just visual inspection — is needed to make fair comparisons between containers; (b) value precision and accuracy in measurement as a professional skill used by Kenyan artisans, jua kali fabricators, and posho mill operators; (c) show curiosity about how shape choice affects storage efficiency.
Key Inquiry Question	How do we determine the volume of solids, and how is volume applied in real-life situations such as grain storage in Eldoret?
Purpose in Storyline	In Lessons 1–4 learners established that surface area describes how much material covers a solid. Now they turn to the inside: How much can the solid hold? By filling and comparing prism and pyramid models with sand, learners discover the factor of ⅓ that links pyramids to prisms of equal base and height, and they compute the exact volume of the posho mill owner's rectangular metal box. This sets up a direct numerical comparison with the cylindrical drum in Lesson 6, moving the class closer to a full explanation of the anchoring phenomenon.
Safety Notes	Handle sand carefully — avoid spilling on the floor to prevent slipping. Ensure transparent containers used for water activities are placed on stable surfaces. If glass containers are used, teacher should supervise pouring. Learners with dust allergies should be seated away from sand-filling activities or offered the water-filling alternative. Clean up all spills immediately.

B. LESSON OVERVIEW

Learners open by revisiting the anchoring phenomenon: the posho mill owner's two metal containers. They predict which holds more grain — the rectangular box or the cylindrical drum — and justify their reasoning using surface area knowledge from Lessons 1–4. In the Observe phase, groups fill rectangular prism, triangular prism, and square-based pyramid cardboard/plastic models with sand (or water), recording volumes by displacement or direct linear measurement. In the Explain phase, the teacher guides derivation of V = base area × height for prisms and V = ⅓ × base area × height for pyramids using the experimental evidence. Learners then compute the volume of the rectangular metal box from the phenomenon using given dimensions, leaving the cylindrical drum as a teaser for Lesson 6. The lesson closes with DQB updates and a model revision where learners annotate their cumulative solid diagrams with volume labels and the derived formulas.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners individually examine the photograph of the posho mill owner's two containers (the rectangular metal box and the cylindrical drum) displayed on the ARES platform or printed sheet. They recall from Lessons 1–4 that both containers were made from the same total area of iron sheet. Each learner writes in their exercise book: (1) a ranked prediction — which container holds more maize grain, or do they hold the same amount? (2) a written justification of 2–3 sentences connecting their prediction to what they already know about surface area and shape. Learners then share predictions with a shoulder partner (Think-Pair) for 2 minutes before three volunteers share with the class. Predictions are recorded on a class tally chart on the board: SAME / RECTANGULAR BOX HOLDS MORE / CYLINDRICAL DRUM HOLDS MORE.	Display the phenomenon photograph and recap with: 'Last lesson we calculated surface area. Today we ask a different question — not how much iron sheet covers these containers, but how much maize grain fits inside them.' Instruct: 'Open your books, write your name, today's date — 2026 — and the title: Volume of Prisms and Pyramids. Now, silently, look at both containers and write your prediction. You have 90 seconds — go.' [WAIT 90 seconds — circulate, read predictions without commenting.] After pair-share: cold-call exactly 3 learners with contrasting predictions — one from each tally column if possible. For each, ask: 'What reasoning connects your prediction to what we learned about surface area?' [WAIT TIME: 12 seconds after each question before accepting answers.] Record tallies visibly. Close with: 'By the end of today you will know the exact answer using mathematics — not guessing.'	Think-Pair-Share with Tally Board: Making predictions public and categorised helps learners confront the intuitive misconception that equal surface area implies equal volume. The tally board creates a shared reference point that the class will return to at the end of the lesson to confirm or revise predictions, making the learning payoff concrete and motivating.	Teacher scans written predictions during the 90-second silent phase. Observable indicator: learner can articulate in writing the distinction between surface area (material used) and volume (space inside). Red flag: if more than 50% predict 'SAME,' teacher notes this and explicitly highlights the surface area vs. volume distinction during the Explain phase.
Observe Phase 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of	Groups of four receive a set of three pre-made cardboard or clear plastic models: (A) a rectangular prism/cuboid (dimensions matching the posho mill box from Lesson 1, scaled down: e.g., 10 cm × 8 cm × 12 cm), (B) a triangular prism (right-angled triangular base, e.g., base legs 6 cm and 8 cm, height 12 cm), and (C) a square-based pyramid with the same base area and height as model A (base 10 cm × 8 cm, height 12 cm — note: for the	Distribute materials and say: 'Your job is to be scientists — measure first, then fill, then record. Do NOT skip the measuring step; the numbers on your data table are the evidence you will use to derive a formula.' Demonstrate the sand-levelling technique once: 'Sweep the ruler flat across the top — like a Kenyan baker levelling a cup of uji flour.' [WAIT for groups to begin.] Circulate with purpose: (1) check that all three dimensions are recorded before any filling begins	Data Aggregation and Pattern Recognition: By posting individual group ratios on a shared class board, learners see that despite small measurement errors, all ratios cluster around ⅓. This 'signal through noise' experience is a key scientific practice (analysing data) and directly motivates the formal derivation in Phase 3. Connecting the levelling step to familiar Kenyan actions (levelling uji flour) reduces procedural anxiety and builds engagement.	Observable indicators: (1) All group data tables have dimensions recorded in three columns (length/base dimensions/height) before any filling — teacher checks this physically during circulation. (2) Group ratio values on the class board fall between 0.28 and 0.38 (acceptable experimental range); outliers flagged for re-measurement. (3) At least one group member can verbally state the pattern ('the pyramid holds about one third of

Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	pyramid, same base and height as the cuboid to make the ⅓ relationship visible). Each model has an open top and a sealed base. Groups first measure and record all dimensions on a provided data table. They then fill model A to the brim with dry sand (or coloured water), level the top with a ruler, and pour the sand into a graduated measuring cylinder or a calibrated transparent container to record the volume in ml/cm^3. They repeat for model B and model C. They record all three volumes in the group data table and compute the ratio Volume(C) ÷ Volume(A) to the nearest tenth. Groups post their ratios on a class data board (sticky note or whiteboard).	— stop groups who rush; (2) ask at least two groups: 'Before you pour, predict the volume of model C compared to model A — same base and height, different shape. Write your prediction in your table.' [WAIT TIME: 10 seconds.] After groups post ratios on the class board, cold-call 2 groups to read their ratio aloud. Ask the class: 'Look at all the ratios on the board — what pattern do you notice?' [WAIT TIME: 15 seconds.] Guide towards: 'They are all close to 0.33 — or one third.'		the prism') when cold-called.
Explain Phase 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class guided derivation using the data from Phase 2. Learners follow in their books, completing a structured notes framework (provided on ARES or as a printed scaffold). Part A — Prism formula: Learners recall that a rectangular prism has a rectangular cross-section. They compute base area ($B = l \times w$) for model A, then observe that Volume = B × h matches the sand data. They confirm with model B using a triangular base ($B = ½ \times$ base × height of triangle). They write and box: $V_{prism} = B \times h$, where B = area of the base. Part B — Pyramid formula: Learners examine the experimental ratio ≈ ⅓ and the teacher presents a brief visual proof using the idea that three identical pyramids fill one prism (shown via diagram or physical demonstration if a set of three pyramid models fitting into	Begin Part A: 'Look at model A data. You measured l = 10 cm, w = 8 cm, h = 12 cm. What is the area of the base?' [WAIT 10 seconds — cold-call 1 learner.] 'Now multiply that by the height. What do you get?' [WAIT 10 seconds.] 'Compare that to your sand measurement. What do you notice?' Proceed to model B: 'What shape is the base of model B? How do we find its area?' [WAIT 12 seconds — cold-call 1 different learner.] For Part B: 'Look at the class board ratios. They cluster around what fraction?' [WAIT 8 seconds.] Draw and explain the three-pyramids-fill-one-prism diagram on the board slowly, asking: 'If three of these pyramids exactly fill this prism, what must the volume of one pyramid equal?' [WAIT 15 seconds — cold-call 1 learner.] Box the formula prominently. For Part C: 'Now let us return to Eldoret. The posho	Structured Notes Framework with Guided Derivation: The scaffolded notes sheet ensures all learners — including those who missed some of Lessons 1–4 — can follow the logical chain from experimental data → pattern → formula → application. The explicit 'box the formula' step signals cognitive closure at each sub-stage. Returning to the Eldoret phenomenon at the computation step anchors abstract formulas in the real-world problem that launched the unit, reinforcing that mathematics is a tool for answering genuine questions.	Observable indicators: (1) Learners' computed volume of the rectangular box = 1.44 m^3 (teacher checks 3–4 books during the 4-minute silent work period). (2) When cold-called, learner can correctly identify $B = l \times w$ for cuboids and $B = ½bh$ for triangular prisms without prompting. (3) Cylinder prediction sentence includes reference to the radius/circular base, indicating awareness that a different base area formula will be needed — this previews Lesson 6.

	the prism is available). Learners write and box: V_pyramid = ⅓ × B × h. Part C — Application to the phenomenon: Using the dimensions of the posho mill owner's rectangular metal box given in Lesson 1 (e.g., length = 1.2 m, width = 0.8 m, height = 1.5 m), learners individually calculate the volume. Teacher then reveals the cylinder dimensions (radius = 0.5 m, height = 1.5 m) and says: 'We will calculate the cylinder's volume next lesson — but can you already predict which is larger?' Learners write a sentence prediction with mathematical reasoning.	mill owner's rectangular box has these dimensions...' Write on board. 'Calculate the volume independently — 4 minutes, silent work.' [WAIT 4 minutes.] Cold-call 1 learner to give the answer and 1 different learner to verify the steps. Then reveal cylinder dimensions and ask: 'Write a one-sentence prediction about the cylinder. Use numbers if you can.' [WAIT 90 seconds.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Bending Light] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Comparing Standard Units of Volume (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes one NEW question that this lesson has raised for them about the posho mill phenomenon on a sticky note or in their DQB section of the ARES platform. Prompt stems are provided: 'I now know… but I still wonder…' or 'If the pyramid holds ⅓ of the prism, what fraction does the cylinder hold compared to…?' Learners also review the DQB cards posted in Lessons 1–4 and place a tick (✓) next to any question they now feel can be answered. Three volunteers read their new questions aloud. The teacher categorises the new questions on the board under two columns: ANSWERED THIS LESSON and COMING IN LESSONS 6–8. The class reads the driving question aloud together: 'Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?'	Say: 'You now know how to calculate volume for prisms and pyramids. But the posho mill owner also has a cylinder — a shape with a curved surface and a circular base. Look at the driving question. Can we answer it fully yet?' [WAIT 10 seconds — expect 'No.'] 'Write your new question now — 90 seconds.' [WAIT 90 seconds.] Cold-call 3 learners to read their questions. As each is read, ask the class: 'Thumbs up if this question connects to the driving question.' Categorise questions visibly. Point to the COMING IN LESSONS 6–8 column: 'These are the investigations that will complete our answer for the posho mill owner in Eldoret. Hold onto your predictions.'	Question Sorting on the DQB: Distinguishing 'answered' from 'still open' questions externalises metacognition — learners see their own growing understanding as a map, not just a list of topics. Sorting new questions into future lessons gives purposeful preview without giving away the answer, maintaining productive intellectual tension around the phenomenon.	Observable indicators: (1) Every learner has at least one new written question — teacher spot-checks 5 books. (2) New questions reference shape-specific vocabulary: 'circle,' 'radius,' 'curved surface,' 'cylinder' — indicating learners are connecting today's prism/pyramid work to the upcoming cylinder lesson. (3) Learners can correctly identify at least 2 DQB cards from earlier lessons as 'answered,' demonstrating retention of prior evidence.

Model Building Phase 📹 **VIDEO:** Video: Bending Light Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative annotated diagram of the two posho mill containers (begun in Lesson 1 and updated each lesson). Today they add: (1) volume labels on the rectangular box using today's computed value ($V = 1.44$ m³), written in a different colour from surface area annotations; (2) the two boxed formulas ($V_{prism} = B \times h$ and $V_{pyramid} = \frac{1}{3} \times B \times h$) in a dedicated 'Formulas Learned So Far' box on their model sheet; (3) a question-mark bubble over the cylindrical drum labelled 'V = ? — Lesson 6'; (4) a one-sentence 'Model Statement' at the bottom of their diagram: 'The rectangular box has a volume of ___ m³. I predict the cylindrical drum has a [larger / smaller / equal] volume because ___.' Learners share their model statement with their shoulder partner and refine if needed.	Distribute or direct learners to retrieve their cumulative model diagrams. Say: 'Add today's evidence in a new colour — this way your diagram becomes a record of your growing understanding, lesson by lesson. Add the volume of the box, both formulas, and the question-mark bubble on the cylinder.' [WAIT 5 minutes for annotation.] Circulate and look specifically for: (a) correct volume value on the box, (b) correct formula notation — especially the ⅓ for pyramid. If a learner writes $V = B \times h$ for the pyramid, quietly ask: 'What did the sand experiment tell us about the pyramid compared to the prism of the same base and height?' [WAIT 10 seconds.] Close: 'Tomorrow — Lesson 6 — we fill in that question mark. By the end of next lesson, you will be able to give the posho mill owner a complete mathematical answer. Pack your model safely.' End with exit task (written in book, 2 minutes): 'A jua kali fabricator in Kamukunji, Nairobi, makes a triangular prism grain scoop with a right-triangle base (legs 9 cm and 12 cm) and length 30 cm. What is its volume?' [Answer: $\frac{1}{2} \times 9 \times 12 \times 30 = 1620$ cm³.]	Cumulative Model Annotation with Colour-Coded Layers: Using a new colour for each lesson's additions transforms the model into a visual learning timeline. The deliberate question-mark bubble over the cylinder makes the incompleteness of the current model visible and motivating — learners leave the lesson with a specific, named gap to fill. The Kamukunji exit task embeds a Kenyan context and provides the teacher with immediate whole-class formative data before the next lesson.	Observable indicators: (1) Exit task answer of 1620 cm³ — teacher collects or photographs 5–6 exit tasks at the door; correct answer confirms formula application. (2) Model statement includes a directional prediction about the cylinder (larger) with a reason referencing circular base area — this shows transfer of thinking beyond today's content. (3) Formula box on diagram shows both formulas with correct notation; the ⅓ coefficient is present in the pyramid formula.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the sand-filling experiment generate ratio values close enough to ⅓ for the pattern to be visible? If groups got widely scattered ratios, consider whether model construction was sufficiently precise — pre-made rigid models are preferable to learner-assembled cardboard nets for this activity. (2) Were learners able to correctly identify the base area formula for the triangular prism without prompting? If many struggled, plan a 5-minute recap of area of triangles at the start of Lesson 6 before introducing the circle area formula. (3) Did learners' DQB new questions reference the cylinder and circular base, or did they remain focused only on prisms and pyramids? Questions that do not reference the cylinder suggest the preview of Lesson 6 was not sufficiently explicit — revisit the cylinder photograph at the start of Lesson 6. (4) How many learners correctly completed the Kamukunji exit task? A success rate below 70% signals that the triangular prism base area ($\frac{1}{2} \times b \times h$) needs re-teaching. (5) Were there learners who remained convinced that equal surface area means equal volume even after Phase 3? These learners need a targeted one-on-one conversation using the numerical results: surface area of the box ≈ surface area of the drum, but volume of the box = 1.44

m^3 while the drum's volume will be shown in Lesson 6 to be larger.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners filled rectangular prism, triangular prism, and square-based pyramid models with sand and measured volumes using graduated containers, then posted group ratio values (pyramid volume ÷ prism volume) on a class data board, observing that all ratios clustered near 0.33.
What did I learn?	Learners derived that the volume of any prism equals base area multiplied by height ($V = B \times h$), and that the volume of a pyramid equals one third of the volume of a prism with the same base and height ($V = \frac{1}{3} \times B \times h$). They computed the volume of the posho mill owner's rectangular metal box as 1.44 m^3 and predicted that the cylindrical drum holds more grain despite being made from the same area of iron sheet.
How does this explain the phenomenon?	Equal surface area does not guarantee equal volume — the shape of the solid determines how much space is enclosed. The rectangular box encloses 1.44 m^3 of grain, but the cylindrical drum (with the same surface area of iron sheet) encloses a larger volume because the circular cross-section is more space-efficient than the rectangular cross-section for a given perimeter, a relationship that will be fully quantified in Lesson 6 when learners derive the volume formula for cylinders.

LESSON 6 (40 minutes): Volume of Cones, Frustums and Spheres — Solving the Posho Mill Mystery	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners derive and apply the volume formulas for cones, frustums, and spheres, then use these formulas — alongside the cylindrical drum volume — to formally and completely resolve why the posho mill owner's cylindrical drum holds more maize grain than the rectangular metal box of equal surface area.
Knowledge	– The volume of a cone is exactly one third of the volume of a cylinder with the same base radius and height: $V = \frac{1}{3}\pi r^2 h$ – The volume of a sphere is $V = \frac{4}{3}\pi r^3$, derivable from the relationship between a hemisphere and a cylinder via Cavalieri's principle – The volume of a frustum is $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$, obtained by subtracting the volume of the removed small cone from the original large cone – Equal surface area does NOT guarantee equal volume — shape geometry determines how efficiently a given area of material encloses space
Skills	– Derive volume formulas for cones and spheres by reasoning from the cone-cylinder pouring relationship and Cavalieri's principle – Apply $V = \frac{1}{3}\pi r^2 h$, $V = \frac{4}{3}\pi r^3$, and $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ to calculate volumes of solids in real-life Kenyan contexts – Compare volumes of solids of equal surface area and draw conclusions about shape efficiency for storage design
Attitudes	– Appreciate that mathematical reasoning — not just intuition — is the tool that resolves the posho mill owner's puzzlement about equal material yet unequal capacity – Value precision and systematic derivation as the foundation of practical decisions made by Kenyan artisans, farmers, and engineers
Key Inquiry Question	How do we determine the volume of cones, frustums, and spheres, and how does knowing these volumes help us explain why the cylindrical drum holds more grain than the rectangular box?
Purpose in Storyline	This lesson completes the volume evidence-gathering sequence by giving learners the cone and sphere volume formulas. With the cylindrical drum volume now calculated and compared directly against the rectangular box volume of 1.44 m³ established in Lesson 5, learners formally close the central mystery of the anchoring phenomenon: equal surface area does not mean equal volume, and the cylinder's geometry makes it the more space-efficient container.
Safety Notes	When using water-pouring demonstration models, ensure containers are placed on a tray or basin to catch spills. Mop up any water on the floor immediately to prevent slipping. If plastic or cardboard models are used, handle cut edges carefully.

B. LESSON OVERVIEW

This lesson is the resolution lesson of the eight-lesson sequence on Surface Area and Volume of Solids. Students arrive knowing the volume of the posho mill owner's rectangular metal box (1.44 m³, from Lesson 5) but they do not yet have a formula for the volume of the cylindrical drum. The lesson opens with a provocative prediction: learners estimate whether the cylindrical drum — made from the same 59,200 cm² of iron sheet calculated in Lesson 1 — holds more or less than 1.44 m³ of maize grain, and by how much. This reactivates the anchoring phenomenon and creates genuine intellectual need for the cone and cylinder volume formulas.

The core evidence-gathering activity uses a physical or teacher-demonstrated water-pouring experiment with matching cone-and-cylinder models (same base radius, same height). Learners observe that it takes exactly three full cones of water to fill the cylinder, establishing $V\_cone = \frac{1}{3}V\_cylinder = \frac{1}{3}\pi r^2 h$ empirically. The teacher then connects this to the sphere via Cavalieri's principle using a visual cross-section argument, deriving $V = \frac{4}{3}\pi r^3$. The frustum formula is derived by subtraction of two cone volumes. With all three formulas in hand, learners calculate the volume of the cylindrical drum (radius ≈ 38.7 cm derived from the known surface area and height from Lesson 1 dimensions, height 100 cm) and compare it directly against 1,440,000 cm^3 for the box.

The lesson culminates in the formal resolution of the driving question: the cylindrical drum holds approximately 1.83 m^3 versus the box's 1.44 m^3 — nearly 27% more grain — from the same sheet of iron. Learners articulate in writing why this is the case (cylinder's circular cross-section encloses more area per unit perimeter than a rectangle, by the isoperimetric principle) and update both the Driving Question Board and their cumulative models. This resolution also launches a forward question: could a sphere of the same surface area hold even more? — positioning Lesson 7 (composite solids and optimisation) as the natural next step.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Writing proportional equations from tables] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Comparing Standard Units of Volume (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a photograph of the two Eldoret posho mill containers side by side — the cylindrical drum and the rectangular metal box — with the caption: 'Both made from 59,200 cm^2 of iron sheet. The box holds 1,440,000 cm^3 of maize. How much does the drum hold?' Each learner independently writes a numerical estimate in their exercise book and a one-sentence justification. They then share estimates with a shoulder partner and note whether they agree or disagree. The class prediction range is recorded on the board.	Display the photograph or sketch of both containers on the board. Say: 'In Lesson 5, we proved that Wanjiku's rectangular metal box holds exactly 1.44 million cubic centimetres of maize grain. Today we have one job — calculate the volume of Omar's cylindrical drum and settle this mystery once and for all.' Pause. Then say: 'Before we calculate anything, I want your best estimate. Write a number in cubic centimetres — higher than the box, lower than the box, or the same? And write ONE reason for your estimate. You have 90 seconds. Go.' [Allow 90 seconds of silent individual writing.] Cold-call 4 students for their estimates — specifically pick one student who predicts lower, one who predicts the same, and two who predict higher. Record all four estimates on the board in a column labelled 'OUR PREDICTIONS'. Say: 'Keep these in view. By the end of this lesson, one of us will be proven right — with mathematics.'	Anchored Prediction with Numerical Commitment — by requiring a specific number (not just 'more' or 'less'), learners activate their existing mental models of how volume relates to shape. Recording predictions publicly creates accountability and intellectual motivation to resolve the discrepancy, directly re-engaging the anchoring phenomenon.	Teacher scans written estimates for the quality of justification. Red flag: learners who write 'the same' because 'same metal sheet = same volume' — these learners still hold the core misconception the lesson must correct. Flag these students for targeted cold-calling during the Explain phase. Green flag: learners who write 'more, because a circle wastes less perimeter' — these show readiness for the isoperimetric extension.
Observe Phase 📹 **VIDEO:** [Writing proportional	The teacher demonstrates the cone-cylinder water-pouring experiment at the front using two transparent matching plastic models (same base radius r = 5	Hold up the cone and cylinder models and say: 'These two shapes have the same base radius — 5 centimetres — and the same height — 12 centimetres. If I fill	Empirical Demonstration with Ratio Reasoning — the physical pouring experiment converts an abstract algebraic relationship ($V\_cone = \frac{1}{3}V\_cylinder$) into a	Teacher checks learners' observation tables: all three entries should read ⅓, ⅔, and 1 (or 3/3). Learners who record decimals without a fraction relationship have

equations from tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	cm, same height h = 12 cm — a cone and a cylinder). Learners observe as the teacher fills the cone with coloured water (water tinted with food dye for visibility) and pours it into the cylinder — once, twice, three times. They record in a table: Pour 1 — fraction of cylinder filled; Pour 2 — fraction filled; Pour 3 — fraction filled. They also watch a 2-minute ARES simulation video clip showing Cavalieri's principle applied to a hemisphere and cylinder (cross-sectional area equality), which provides the visual justification for $V\_sphere = \frac{4}{3}\pi r^3$.	this cone with water and pour it into the cylinder, what fraction of the cylinder do you think will be filled? Thumbs up if you think more than half, thumbs sideways if you think exactly half, thumbs down if you think less than half.' [Wait 10 seconds, scan the room.] 'Watch carefully and count with me.' Perform Pour 1 — ask: 'What fraction is filled?' [Expected: one third.] Confirm on board. Perform Pour 2 — ask: 'Now?' [Two thirds.] Pour 3 — ask: 'Now?' [Full.] Say: 'Say the relationship aloud with me — three cones fill one cylinder. So the volume of the cone is…?' [Wait 10 seconds for silent processing.] Cold-call 2 students. Write $V\_cone = \frac{1}{3} \times \pi r^2 h$ on board. Then say: 'Now for the sphere — we cannot do a simple pouring experiment, but watch this video clip. Your job: in the video, what two shapes are being compared, and why does that comparison give us the sphere formula?' Play the Cavalieri clip. After clip, cold-call 3 students to narrate what they observed.	directly observed counting fact (3 cones = 1 cylinder). This is a classic NGSS Science and Engineering Practice 3 (Planning and Carrying Out Investigations) applied to a deductive mathematical context. The video extends this to the sphere via visual cross-sectional argument, bridging empirical and theoretical reasoning.	not yet made the ratio connection — prompt them with 'How many cones fit in one cylinder? Write that as a fraction.' For the Cavalieri video: ask learners to sketch the cross-sectional comparison from memory; learners who can draw the hemisphere and cylinder with equal cross-sections at height h demonstrate conceptual uptake.
Explain Phase 📹 **VIDEO:** Writing proportional equations from tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Working in pairs, learners complete a structured derivation-and-application worksheet with four tasks: (1) Write the cone volume formula and use it to find the volume of a conical maize storage bin used at a farm in Kericho — radius 0.6 m, height 1.2 m. (2) Use the sphere formula $V = \frac{4}{3}\pi r^3$ to find the volume of a spherical water tank at a homestead in Kisumu — radius 0.9 m. (3) Derive the frustum formula by subtracting cone volumes — given a frustum with R = 7 cm, r = 4 cm, h = 9 cm (slant height known from Lesson 3). (4) The critical task: use the surface	Distribute the worksheet. Say: 'Tasks 1, 2, and 3 are practice — you have seen these formulas. Task 4 is the mathematics that settles our lesson-one mystery. Work in pairs. You have 12 minutes.' [Circulate actively.] At the 6-minute mark, pause the class and cold-call a pair on Task 1 to narrate their working aloud — correct any errors publicly. At the 10-minute mark, cold-call a different pair on Task 4 to share the radius they calculated. Say: 'What radius did you get for the cylindrical drum?' [Expected: from $2\pi r^2 + 2\pi r(100) = 59{,}200$, solving gives $r \approx 38.7$ cm.] Write this on	Backward Problem-Solving (Working from Surface Area to Volume) — Task 4 reverses the typical direction of calculation. Learners must first extract the radius from the known surface area before they can compute the volume. This integrates learning from Lessons 1 and 3 (surface area formulas) with today's volume formula, making explicit that surface area and volume are linked by geometry but are distinct quantities. This is NGSS Practice 5: Using Mathematics and Computational Thinking.	Targeted check on Task 4: (a) Does the pair correctly set up $2\pi r^2 + 2\pi rh = 59{,}200$ with h = 100? (b) Do they solve the quadratic correctly for $r \approx 38.7$ cm? (c) Is their final volume calculation correct and expressed in cm^3? (d) Is their written conclusion specific — citing the actual numbers — rather than vague ('the drum is bigger')? Flag pairs whose conclusion says 'because the cylinder is rounder' without a mathematical explanation for targeted questioning during DQB phase.

	area of Omar's cylindrical drum (59,200 cm², height h = 100 cm from Lesson 1) to back-calculate the radius using SA_cylinder = $2\pi r^2 + 2\pi rh$, then compute the drum's volume $V = \pi r^2 h$. Compare this volume with Wanjiku's box volume of 1,440,000 cm³ and write a conclusion.	the board. Then say: 'Now put that radius into $V = \pi r^2 h$. What do you get?' [Expected: $V \approx \pi \times (38.7)^2 \times 100 \approx 471{,}969\pi \approx 1{,}483{,}000$ cm³ … teacher note: actual answer depends on exact r; use r ≈ 38.7 cm giving V ≈ 1,483,169 cm³ ≈ 1.483 m³. Teacher should also note the worked SA back-calculation: $2\pi r^2 + 200\pi r = 59200 \rightarrow r^2 + 100r - 9431 \approx 0 \rightarrow r \approx 38.7$ cm.] Write both volumes side by side: DRUM ≈ 1,483,000 cm³; BOX = 1,440,000 cm³. Say: 'The drum holds about 43,000 cm³ — or 43 litres — MORE than the box. From identical iron sheet. Discuss with your partner: WHY? You have 60 seconds.' [Wait 60 seconds.] Cold-call 3 pairs for their explanations.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Writing proportional equations from tables Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Comparing Standard Units of Volume (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front of the room. Learners review the questions posted in earlier lessons and identify which questions this lesson has now answered. Each learner writes one sticky note with a RESOLVED claim — a sentence beginning 'We can now explain that…' — and one new question that this lesson raises but has not yet answered (e.g., 'Would a sphere of the same surface area hold even more?' or 'Is there a formula to find the shape that holds the most for a given surface area?'). New questions are added to the board under the heading 'Still Wondering'.	Point to the DQB and say: 'Look at the questions we have been building since Lesson 1. Find any question that begins with the words same material, different volume, or why does the drum hold more. Today's mathematics has given us the answer.' [Wait 15 seconds for learners to scan the board.] Say: 'Write your RESOLVED note — start with: We can now explain that… — and your new wonder question. You have 3 minutes.' [Allow 3 minutes.] Invite 4 learners to read their RESOLVED notes aloud — cold-call across ability levels. After each one, say: 'Does everyone agree this is now resolved? Thumbs up if yes.' Then say: 'Now read your new wonder questions.' Accept 3–4 new questions and physically add them to the 'Still Wondering' section of the board. Say: 'These questions are exactly what Lessons 7 and 8 will tackle — composite solids and which shape	Claim-Evidence-Reasoning (CER) on the DQB — the RESOLVED note format (We can now explain that…) pushes learners to state a claim supported by the mathematical evidence gathered in this lesson. This public documentation of growing understanding mirrors how scientists update models as new evidence accumulates, directly connecting to NGSS Practice 6: Constructing Explanations.	Teacher reads RESOLVED notes for two criteria: (1) Does the claim include a specific numerical comparison (e.g., 'the drum holds 1,483,000 cm³ versus the box's 1,440,000 cm³')? (2) Does the claim name the mathematical reason (same surface area but different shape)? Vague claims ('the drum is better') indicate the learner has the intuition but not the mathematical articulation — these learners need the Model Building phase to consolidate.

		is most efficient. You are asking the right questions.'		
Model Building Phase 📹 **VIDEO:** [Writing proportional equations from tables] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Comparing Standard Units of Volume (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative 'Posho Mill Containers' model diagram — a cross-sectional sketch of both containers with annotated measurements and formulas, begun in Lesson 1 and updated each lesson. Today they add three items to their model: (1) the cone-cylinder volume relationship arrow labelled V_cone = ⅓V_cylinder, illustrated with a small sketch of the pouring experiment; (2) the annotated volume calculation for the cylindrical drum, showing the back-calculation of radius from surface area and the final volume of ≈ 1,483,000 cm^3; (3) a comparison bar or arrow between the two containers labelled 'DRUM holds ~43,000 cm^3 MORE than BOX — same iron sheet used'. Learners also write a one-paragraph 'Advice to the Artisan' at the bottom of their model, addressed to the metalsmith in Eldoret, explaining in plain language why the cylindrical drum is the more efficient grain storage shape and whether the artisan should always build cylinders.	Say: 'Take out your cumulative model from your portfolio. You have been building this since Lesson 1. Today we close the central mystery — update your model with the three things on the board.' [Write on board: (1) Cone-cylinder relationship sketch and formula; (2) Drum radius back-calculation and volume = _____ cm^3; (3) Comparison label showing which holds more and by how much.] 'You also have a creative task: write three to five sentences of advice to the metalsmith — Omar Khalif — who built these containers. What would you tell him? Is cylindrical always better? What about a sphere? You have 6 minutes.' [Circulate and read 'Advice to the Artisan' paragraphs as learners write.] After 6 minutes, cold-call 2 learners to read their advice paragraphs aloud. After each, ask the class: 'What did this learner say that was mathematically strong? What could be stronger?' [Wait 10 seconds between each question.] Close by saying: 'You have now proven, with mathematics, what the posho mill owner could only sense by pouring grain. Next lesson, we will find out whether a sphere could be even better — and whether there is a perfect shape for storage.'	Cumulative Model Annotation with Expert Communication (Advice to the Artisan) — updating the running diagram builds a visible, personal record of the entire evidence sequence, making the storyline arc tangible. The 'Advice to the Artisan' paragraph demands that learners translate mathematical conclusions into practical guidance, which is NGSS Practice 8: Obtaining, Evaluating, and Communicating Information, and directly mirrors the real-world purpose of the driving question.	Collect and spot-check 6 model diagrams (2 high-achieving, 2 mid, 2 struggling learners). Check: (a) Is the cone-cylinder ratio correctly shown as 1:3? (b) Is the drum radius calculation shown as a quadratic solved from SA = 59,200 cm^2? (c) Is the volume comparison labelled with actual numbers? (d) Does the 'Advice to the Artisan' paragraph go beyond 'use a cylinder' and engage with the mathematical reason (circular cross-section encloses more area per perimeter unit)? Use findings to plan Lesson 7 entry point.

D. TEACHER REFLECTION

1. During the Predict Phase, how many learners still held the misconception that equal surface area implies equal volume? Did the final resolution in the Explain Phase visibly shift those learners' thinking — and how do you know (what did you observe or hear)?

2. The cone-cylinder pouring demonstration is the empirical heart of this lesson. Was the demonstration clear enough for learners in the back rows to observe the three pours and the fractions? If you teach this lesson again, what modification to the setup (container size, food-dye colour, positioning) would improve visibility?

3. Task 4 on the worksheet (back-calculating radius from surface area) requires learners to solve a quadratic equation — a skill from Strand 1. How many pairs successfully set up $2\pi r^2 + 2\pi rh = 59{,}200$ and solved it correctly? What scaffolding (hint card, partially completed working) would support struggling pairs without removing the productive struggle?

4. The 'Advice to the Artisan' paragraph is the first time in this sequence that learners are asked to communicate mathematics to a non-mathematician. Did learners write in plain language with mathematical evidence, or did they default to formula-copying? What sentence starter or model paragraph would help learners bridge the gap between mathematical notation and practical communication?

5. The lesson ends with an open forward question: could a sphere of the same surface area hold even more? Did learners generate this question themselves in the DQB phase, or did it need teacher prompting? What does this tell you about learners' readiness to think beyond the given problem and engage with optimisation ideas in Lesson 7?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a physical water-pouring demonstration in which a cone and a cylinder of identical base radius and height were used: it took exactly three full cones of water to completely fill the cylinder, establishing the 1:3 volume ratio empirically. They also watched a visual simulation of Cavalieri's principle showing equal cross-sectional areas between a hemisphere and a cylinder at every height.
What did I learn?	The volume of a cone is $V = \frac{1}{3}\pi r^2 h$ — one third of the volume of a cylinder with the same base and height. The volume of a sphere is $V = \frac{4}{3}\pi r^3$, derived using Cavalieri's cross-sectional argument. The volume of a frustum is $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$. Using these formulas, learners back-calculated the cylindrical drum's radius from its known surface area of 59,200 cm^2, found $r \approx 38.7$ cm, and computed the drum's volume as approximately 1,483,000 cm^3 — compared to the rectangular box's volume of 1,440,000 cm^3.
How does this explain the phenomenon?	The posho mill mystery is now formally resolved: the cylindrical drum holds approximately 43,000 cm^3 — about 43 litres — more maize grain than the rectangular metal box, even though both were made from exactly the same area of iron sheet. This is because a circular cross-section encloses a greater area for a given perimeter than a rectangular cross-section does. Equal surface area does not guarantee equal volume — the geometry of the shape determines how efficiently a given amount of material encloses space. This is a principle that Kenyan artisans, grain store designers, and engineers can use to choose the most efficient container shape for a given amount of construction material.

LESSON 7 (80 minutes): Volume and Surface Area of Composite Solids & Real-Life Applications	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to calculate the total surface area and volume of composite solids and apply these calculations to real-life Kenyan contexts such as water storage tanks and grain silos.
Knowledge	Learners will know how to identify and decompose composite solids (e.g., a cylinder topped with a hemisphere, a cone mounted on a cylinder) into their component shapes, and apply combined formulae to determine total surface area and total volume.
Skills	Learners will be able to: (a) decompose a composite solid into recognisable component solids; (b) calculate the total surface area of composite solids by summing the exposed surfaces of each component; (c) calculate the total volume of composite solids by summing the volumes of each component; (d) apply these calculations to multi-step real-life problems involving material cost (surface area) and storage capacity (volume).
Attitudes	Learners will appreciate how mathematics empowers Kenyan artisans, engineers, and farmers to make cost-effective and practical decisions in construction and storage design — linking back to the posho mill owner's puzzle about the cylindrical drum versus the rectangular box.
Key Inquiry Question	How do we determine the surface area and volume of solids, and how are surface area and volume of solids applied in real-life situations?
Purpose in Storyline	In Lesson 7, learners move from calculating surface area and volume of individual solids (Lessons 1–6) to tackling composite solids — shapes that are combinations of two or more solids. By solving a problem about a cylindrical water tank with a hemispherical top (common in Kenyan water-harvesting projects in Makueni and Kitui counties) and a grain silo problem set in the Rift Valley, learners synthesise all formulae learned in the sequence. This directly advances the driving question: the posho mill owner's artisan must now design a composite storage container that maximises volume for a given sheet metal area, requiring mastery of both surface area and volume of composite solids.
Safety Notes	No hazardous materials are used in this lesson. If physical composite solid models are handled, remind learners to be careful with sharp edges on metal or cardboard models. Ensure adequate space for group work to avoid tripping on bags or chairs.

B. LESSON OVERVIEW

Learners begin by predicting the volume and surface area of a composite solid (a cylinder topped with a hemisphere — the shape of a common Kenyan water tank) before any instruction. They then observe a worked decomposition of the composite solid on the board alongside a physical or printed model, gathering evidence by measuring and calculating each component. In the Explain phase, groups apply their understanding to a structured Rift Valley grain silo problem (a cone on top of a cylinder) and present solutions, making the multi-step reasoning explicit. The Driving Question Board is updated to capture the insight that composite solids require combining formulae and careful identification of which surfaces are internal (not exposed). Finally, learners revise their cumulative model — the annotated diagram of the posho mill owner's containers — to now include a composite design that could outperform both the cylindrical drum and the rectangular box from the original phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Global winds and currents Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prime and Composite Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a labelled diagram of a composite solid: a closed cylinder of radius 0.7 m and height 1.5 m, topped with a solid hemisphere of the same radius 0.7 m — the shape of a water tank commonly installed in water-harvesting projects in Makueni County. The composite solid is labelled 'Mwangi wa Kamau Water Tank.' Individually (no calculators yet), learners write in their exercise books: (1) a rough sketch of how they would split the shape into known solids; (2) a prediction — do they expect the total surface area to be more or less than a complete closed cylinder of the same radius and height 1.5 m + 0.7 m = 2.2 m? Why? (3) a prediction of which component — the cylindrical body or the hemispherical top — contributes more to the total volume. Learners share predictions with a seat partner (Think-Pair) for 2 minutes before the teacher takes cold-call responses.	Display the diagram of the Mwangi wa Kamau Water Tank on the board (or ARES screen). Say: 'Before we calculate anything, I want you to look at this tank. In your books, sketch how you would break it into shapes you already know. Then answer the two prediction questions on the board. You have 4 minutes — work alone.' Set a visible 4-minute timer. Circulate silently; note students who split correctly (cylinder + hemisphere) versus those who treat the whole as a cylinder. After 4 minutes say: 'Turn to your neighbour and share your sketch and predictions for 2 minutes.' After pair-share, cold-call 3 students: one who split correctly, one who is unsure, one who predicted volume. Ask each: 'Tell us how you split it and what you predicted — and why.' Apply 10-second WAIT TIME after each question before accepting the response. Do NOT correct predictions yet. Say: 'Hold your predictions — we are going to find out if the mathematics agrees with you.'	Think-Pair-Predict: This strategy activates prior knowledge of individual solids (cylinder, hemisphere) and forces learners to confront the central challenge of composite solids — decomposition. By committing a prediction to paper before instruction, learners create a cognitive anchor that makes the subsequent evidence phase more meaningful. Connecting to the phenomenon: just as the posho mill owner had to look at two containers and guess which held more, learners are practising the same intuitive reasoning before applying rigorous mathematics.	Observable indicator: Can the learner produce a sketch that correctly identifies the cylinder and hemisphere as the two components? Listen during cold-call for whether learners spontaneously mention 'the flat circle where they join is an internal surface' — this reveals whether they are already thinking about exposed versus hidden surfaces, the key conceptual challenge of this lesson. Note names of learners who struggle with decomposition for targeted support in Phase 2.
Observe Phase 📹 **VIDEO:** Global winds and currents Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prime and Composite Numbers	The teacher leads a structured decomposition demonstration on the board. The Mwangi wa Kamau Water Tank (radius r = 0.7 m, cylinder height h = 1.5 m, hemisphere radius r = 0.7 m) is decomposed step by step. Learners complete a structured observation table in their books as the teacher works through each component: COMPONENT | FORMULA |	Say: 'We are going to decompose the Mwangi wa Kamau Water Tank together. I need every learner filling in this table as we go — do not just copy; substitute the numbers yourself.' Write the observation table on the board column by column. At each 'EXCLUDED' surface, pause and ask: 'Why do we NOT include this circle in the surface area?' Apply 12-second WAIT TIME. Cold-call 2 students. Emphasise: 'This circle is glued inside — grain or water	Structured Observation Table with Decomposition: By requiring learners to fill in each row of a table rather than passively watching, this strategy makes the reasoning process visible and auditable. The explicit 'EXCLUDED' row for internal surfaces is the key sensemaking move — it directly addresses the misconception that surface area of a composite solid equals the simple sum of the full surface areas of its parts. This connects to	Observable indicator: After the second worked example, learners' tables should show the correct exclusion of the shared circular face between the cone base and cylinder top. Circulate and check: (1) Is the learner using slant height l (not perpendicular height) for the cone's lateral surface? (2) Has the learner excluded the top circle of the cylinder (shared with cone base) from the surface area? (3) Is the total volume the sum of cone + cylinder volumes (not difference)?

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	SUBSTITUTION | VALUE Cylindrical curved surface | 2πrh | 2 × π × 0.7 × 1.5 | ___ Circular base of cylinder | πr² | π × 0.7² | ___ Top circle of cylinder (INTERNAL — covered by hemisphere base) | πr² | EXCLUDED | × Curved surface of hemisphere | 2πr² | 2 × π × 0.7² | ___ Flat base of hemisphere (INTERNAL — fused to cylinder top) | πr² | EXCLUDED | × TOTAL SURFACE AREA | Sum of exposed surfaces | | ___ Then volume: COMPONENT | FORMULA | SUBSTITUTION | VALUE Volume of cylinder | πr²h | π × 0.7² × 1.5 | ___ Volume of hemisphere | (2/3)πr³ | (2/3) × π × 0.7³ | ___ TOTAL VOLUME | Sum | | ___ Learners check their predictions against the calculated values and annotate their prediction sketches with tick or cross. A second worked example is shown: a conical cap (radius 0.5 m, slant height 0.8 m) placed on top of a closed cylinder (radius 0.5 m, height 1.2 m) — resembling a grain silo roof in Nakuru. Learners observe the teacher decompose and identify internal surfaces, then attempt the surface area and volume calculations in their books while the teacher circulates.	cannot touch it, a painter cannot paint it. It is internal.' After completing the first worked example, say: 'Check your prediction. Thumbs up if the total volume was bigger than your prediction, thumbs sideways if about right, thumbs down if smaller.' Scan the room. Say: 'Interesting — let us remember this when we update our Driving Question Board.' For the second worked example (conical silo), say: 'I will decompose it for you — you fill in the table. Then for 5 minutes, work out the surface area and volume on your own. I will cold-call three groups to share one step each.' Circulate and identify calculation errors (especially forgetting to use slant height l for cone lateral surface area πrl versus radius for base). After 5 minutes cold-call 3 students to share one row of their table each, correcting errors publicly and affirmatively: 'Good attempt — the slant height l = 0.8 m is used here, not the perpendicular height. Let us fix that together.'	the phenomenon: the posho mill owner's metalsmith must know exactly how much iron sheet is exposed (to rust, to paint, to cost) — not the internal joints.	Flag any learner who includes the internal circle or uses the wrong height for the cone lateral surface — provide immediate targeted correction before Phase 3.
Explain Phase 📹 **VIDEO:** Global winds and currents	Learners work in groups of 4 on the 'Rift Valley Grain Silo Challenge' — a structured multi-step problem card:	Distribute problem cards. Say: 'This is the Rift Valley Grain Silo Challenge. You have 18 minutes. Every member must be writing —	Collaborative Problem Solving with Role Differentiation and Gallery Walk: Assigning Calculator, Recorder, Checker, and Presenter	Observable indicators during group work: (1) Does the group correctly compute slant height using Pythagoras before

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prime and Composite Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	'Mwalimu Chebet, a farmer in Eldoret, wants to build a grain silo for storing maize after harvest. The silo consists of a cylinder of radius 1.2 m and height 3.5 m, topped with a conical roof of the same radius and perpendicular height 0.9 m. (a) Calculate the slant height of the conical roof. (b) Calculate the total surface area of the silo (the base of the cylinder is concrete — exclude it; the internal circle between cylinder and cone is not painted). (c) Calculate the total volume of grain the silo can hold. (d) Iron sheet costs Ksh 850 per square metre. Calculate the cost of iron sheet needed to make the curved surfaces and conical roof of the silo. (e) Mwalimu Chebet's neighbour, Wanjiku, has a silo of the same total height (4.4 m) but is a pure cylinder of radius 1.2 m. Compare the volume of the two silos. Which design is better for storage? Explain your answer mathematically.' Each group assigns roles: Calculator, Recorder, Checker, Presenter. Groups work for 18 minutes. Each group writes their solution on a manila paper / whiteboard section. The Presenter from each group shares part (c) and (e) with the class. The class uses a gallery walk (2 minutes) to view other groups' solutions before the teacher consolidates.	Calculator works out numbers, Recorder writes on manila paper, Checker verifies each step, Presenter prepares to explain part (c) and (e) to the class. Start with part (a) — you need the slant height before anything else.' Set a visible 18-minute timer. At the 5-minute mark, circulate and whisper to groups still on part (a): 'Pythagoras — $l^2 = r^2 + h^2$; you have r = 1.2 and h = 0.9.' At the 12-minute mark, prompt groups finishing early: 'Have you compared volumes in part (e)? What does the extra 0.9 m of height as a cone give you versus 0.9 m as a cylinder?' After 18 minutes, say: 'Presenters, you have 90 seconds to explain your answer to parts (c) and (e) — one group at a time.' After all presentations, conduct a 2-minute gallery walk. Then consolidate on the board: write the correct solution to all five parts. Ask the class: 'In part (e), which holds more — the composite silo or the pure cylinder? Why? What does this remind you of?' Apply 10-second WAIT TIME. Cold-call 2 students. Guide toward: 'A cone on top holds LESS than a cylinder of the same height — so if Mwalimu Chebet wants maximum storage, she might prefer the pure cylinder, but the conical roof sheds rainwater better.' Say: 'This is real engineering — there is a trade-off between volume and function, just like the posho mill owner's containers.'	roles ensures all learners are cognitively active and accountable. The multi-part problem card scaffolds complexity: parts (a)–(c) are procedural, part (d) introduces cost (real-life application), and part (e) is comparative and evaluative — requiring learners to connect surface area and volume to engineering decisions. The gallery walk exposes learners to multiple solution approaches and common errors simultaneously. The explicit comparison in part (e) directly echoes the driving question phenomenon: same 'footprint,' different shape, different volume.	proceeding? $l = \sqrt{1.2^2 + 0.9^2} = \sqrt{1.44 + 0.81} = \sqrt{2.25} = 1.5$ m. (2) Does the group exclude the base circle and the internal shared circle correctly in the surface area? (3) Is the cost calculation in part (d) based on the correct surface area (curved cylinder + cone lateral surface only)? (4) In part (e), does the group correctly compute the volume of the pure cylinder as $\pi \times 1.2^2 \times 4.4$ and compare it to the composite silo volume, and draw a valid conclusion? During presentations, listen for whether the Presenter can explain WHY the internal surface is excluded — this is the target conceptual understanding for this lesson.
Driving Question	Each learner receives two sticky notes. On the first (green), they	Distribute sticky notes. Say: 'You have 3 minutes. On the green	Driving Question Board as a Living Conceptual Map: The two-column	Observable indicator: Read green notes for evidence that learners

Board (DQB) Creation 📹 **VIDEO:** Global winds and currents Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prime and Composite Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	write one new thing they can now explain about the posho mill owner's problem that they could not explain at the start of the sequence — specifically connecting composite solids to the phenomenon. On the second (orange), they write one remaining question or a new question prompted by today's lesson (e.g., 'What if the silo had a frustum instead of a cone on top?' or 'How does the artisan decide which composite shape to use?'). Sticky notes are placed on the class DQB under two columns: 'WE CAN NOW EXPLAIN' and 'WE STILL WONDER.' The teacher reads three green notes aloud and three orange notes aloud, then draws a red arrow on the DQB from the original phenomenon question ('Why does the cylinder hold more than the box?') to a new node: 'Composite solids: shape of EACH PART affects total volume AND total surface area independently.' The teacher highlights that the orange questions — especially about optimisation — will be addressed in Lesson 8 (the culminating design task).	note, finish this sentence: I can now explain… connecting it to the posho mill owner's problem. On the orange note, write a new question this lesson has given you.' After 3 minutes, say: 'Come place your notes on the DQB — green in the left column, orange in the right.' As learners place notes, read three green notes aloud, affirming each: 'This is exactly the kind of thinking an engineer uses.' Read three orange notes aloud and say: 'These orange questions are brilliant — especially the ones about choosing the best shape. That is the heart of Lesson 8, our final lesson where you will become the artisan.' Draw the red arrow on the DQB and annotate it: 'Composite solids: surface area ≠ sum of all surfaces; volume = sum of all volumes.' Point to the original phenomenon: 'We started with the mill owner's puzzle. Now we can solve far more complex versions of that puzzle. Lesson 8 — you design the optimal container.'	DQB update makes metacognitive progress visible and public. Green notes reinforce consolidated understanding; orange notes sustain intellectual curiosity and signal readiness for Lesson 8. Drawing a red arrow physically connecting today's learning to the original phenomenon reinforces the storyline coherence — learners see their knowledge growing lesson by lesson toward answering the central driving question. The public reading of notes validates individual thinking and builds classroom community.	can explicitly connect composite solid calculations to the phenomenon (e.g., 'I can now explain that the mill owner could use a composite tank — cylinder plus hemisphere — to store even more grain than just a cylinder, using nearly the same metal sheet'). Flag learners who write green notes that are purely procedural ('I can now calculate surface area of composite solids') without connecting to the phenomenon — these learners need prompting in Phase 5 to make the connection explicit.
Model Building Phase 📹 **VIDEO:** Global winds and currents Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Prime and Composite Numbers	Learners return to their cumulative annotated diagram of the posho mill owner's two containers — the tall cylindrical drum and the rectangular metal box — which they have been building since Lesson 1. In this final model revision before the Lesson 8 design task, learners add a third container to their diagram: a composite solid of their own design (cylinder + hemisphere, or cylinder + cone, or cylinder + cone frustum) with the constraint that its total surface area must equal the	Say: 'Open your cumulative model diagram. You have been building this since Lesson 1. Today you add one final container — a composite solid of your own design. The only rule is: its total surface area must be approximately 4.8 m² — the same metal sheet the mill owner used for his cylindrical drum. You choose the composite shape. You have 10 minutes.' Circulate actively. For learners who are stuck choosing dimensions, say: 'Start with a cylinder of radius r — write surface	Cumulative Model Revision with Design Constraint: Adding a self-designed composite container to the cumulative model under a real constraint (surface area ≤ 4.8 m²) transforms model building from a recording activity into a genuine design task. This previews Lesson 8 and gives learners agency. The constraint replicates the real-world condition of the phenomenon — the metalsmith has a fixed amount of iron sheet — and forces learners to use both surface area (to satisfy the constraint) and	Observable indicators: (1) Has the learner chosen a composite solid and correctly identified all exposed surfaces for the surface area calculation? (2) Is the surface area calculation shown symbolically before numbers are substituted? (3) Does the total surface area approximately equal 4.8 m²? (4) Is the total volume calculated and compared explicitly to the cylindrical drum? Collect the cumulative model diagrams at the end of the lesson for written formative feedback — focus

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	surface area of the original cylindrical drum (given as approximately 4.8 m²). Each learner must: (1) choose their composite solid and label its dimensions; (2) write the formula for its total surface area and show that it equals ≈ 4.8 m²; (3) calculate its total volume; (4) annotate the diagram with an arrow: 'My composite container holds ___ m³ — more / less / same as the cylindrical drum (___ m³).' Learners who finish early extend by sketching how a Rift Valley metalsmith (fundi) would cut and weld the iron sheet pieces to make their composite container.	area formula, set it equal to 4.8 minus the hemisphere surface, then solve for what radius and height are possible.' For learners who finish early, say: 'Sketch how the fundi would cut the iron sheet — what shapes would they cut out?' In the final 3 minutes, ask 2 volunteers to hold up their diagrams and describe their composite container in one sentence. Close the lesson by saying: 'In Lesson 8, you become the artisan. You will use everything in this model to design the optimal storage container for the posho mill owner — and justify your design with mathematics. Keep this model safe.'	volume (to evaluate storage capacity) simultaneously. The extension task (cutting plan for the fundi) integrates creativity and connects to the artisan context in the driving question.	comments on whether the learner has correctly identified internal (excluded) surfaces and whether their volume comparison statement is mathematically justified. Return with feedback before Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the Observe phase, how many learners independently identified the internal shared surface (the circle between the cylinder and hemisphere/cone) as excluded from surface area — without being told? This is the key conceptual hurdle of the lesson; if fewer than 60% identified it spontaneously, consider starting Lesson 8 with a brief re-teaching moment using a physical model (cut a cardboard cylinder and hemisphere apart to show the hidden circle). (2) In the Explain phase, did all groups successfully compute the slant height in part (a) of the Grain Silo Challenge before proceeding? Groups that struggled here reveal a gap in Pythagoras application from earlier strands — note these groups for differentiated support in Lesson 8. (3) Were learners able to articulate the trade-off in part (e) — that a conical top reduces volume compared to a cylindrical extension of the same height, but serves a functional purpose (rainwater shedding)? If this engineering reasoning was absent from presentations, make it the opening discussion of Lesson 8. (4) In the Model Building phase, did learners' self-designed composite containers reflect genuine creative use of the formulae, or did most default to the same shape modelled in class? If creativity was limited, consider providing a 'menu' of three composite solid options in Lesson 8 to scaffold choice. (5) Review the cumulative model diagrams: are learners' annotations growing in mathematical precision across the 7 lessons? Use any regression in quality (e.g., missing units, unsupported volume comparisons) as a signal to revisit precision expectations in the Lesson 8 design task rubric.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed the step-by-step decomposition of two composite solids — a cylinder topped with a hemisphere (Mwangi wa Kamau Water Tank, Makueni County) and a cylinder topped with a cone (Nakuru grain silo) — and completed structured observation tables identifying which surfaces are exposed (included in surface area) and which are internal (excluded), as well as calculating total volumes by summing component volumes.
What did I learn?	Learners learned that: (1) a composite solid is decomposed into component solids for calculation; (2) the total surface area of a composite solid is the sum of only the EXPOSED surfaces — internal surfaces where two components are fused are excluded; (3) the total volume of a composite solid is the sum of the volumes of all component solids; (4) the same total surface area (same

	amount of metal sheet) can yield very different total volumes depending on which composite shape is chosen — directly explaining the posho mill owner's puzzle at a more advanced level.
How does this explain the phenomenon?	Learners explained their understanding by: (1) solving the five-part Rift Valley Grain Silo Challenge (Mwalimu Chebet's silo in Eldoret) in groups, including cost calculations and a comparative volume analysis; (2) updating the Driving Question Board with explicit connections between composite solid calculations and the original phenomenon; (3) designing their own composite storage container with a fixed surface area constraint ($\approx 4.8\ m^2$) and annotating their cumulative model diagram with the resulting volume, comparing it to the posho mill owner's cylindrical drum and rectangular box.

LESSON 8 (80 minutes): Project Presentations, Final Model Building & Unit Consolidation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and demonstrate mastery of surface area and volume of solids by presenting community-based container designs, resolving the anchoring phenomenon, and completing the cumulative unit model.
Knowledge	Learners recall and synthesise surface area and volume formulae for prisms, cylinders, pyramids, cones, frustums, spheres, and composite solids, and articulate why equal surface area does not guarantee equal volume across different solid shapes.
Skills	Learners calculate, compare, and critique surface area and volume values in real-world container designs; construct a final annotated model linking all unit formulae to the posho mill phenomenon; and communicate mathematical reasoning clearly in a group presentation.
Attitudes	Learners appreciate the practical value of geometry in Kenyan artisan trades, construction, and agriculture; demonstrate integrity in accurate measurement and computation; and show respect for peers' design decisions during critique.
Key Inquiry Question	How do we determine the surface area and volume of solids, and how are surface area and volume of solids applied in real-life situations?
Purpose in Storyline	This final lesson brings the unit's storyline to a close. Groups share the community storage container designs they developed as their home/extended project — grain silos, water tanks, or cooking pots for Kenyan community needs — specifying dimensions, total material (surface area) required, and holding capacity (volume). Through structured peer critique, the class surfaces the core insight that has threaded through all eight lessons: shape determines the ratio of capacity to material. The Driving Question Board is completed with a full class-constructed explanation that resolves the posho mill owner's puzzle in Eldoret. Careers that depend on these concepts are mapped, and learners leave with a coherent, consolidated mental model of the entire sub-strand.
Safety Notes	No hazardous materials are used. Ensure physical models or cardboard prototypes are handled carefully to avoid cuts from sharp edges. If scissors or craft knives were used during project construction, confirm they are stored away before presentations begin. Maintain orderly movement when groups display their designs around the room.

B. LESSON OVERVIEW

Lesson 8 is the capstone of Sub-Strand 2.7. In a double-period structure, learner groups (3–4 members) present their home/extended project: a mathematically justified design for a community storage container (grain silo, water tank, or cooking pot) specifying solid shape, labelled dimensions, total surface area of iron sheet or material required, and volume/capacity. Each group receives structured peer critique using a gallery-walk protocol. The teacher then facilitates a whole-class Final Explanation discussion that fully resolves the driving question — why the cylindrical drum at the Eldoret posho mill holds more maize than the rectangular box made from the same sheet of metal. The class completes the Driving Question Board by posting final explanations against every question raised across the unit. Learners compare their Lesson 1 initial model with today's final annotated model, articulating how their understanding grew. The lesson closes with a mapping of careers — metalsmith, civil engineer, packaging designer, water-tank fabricator, KEBS standards officer — that depend on surface area and volume mathematics.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners individually re-read the	Display the phenomenon	Predict-Revisit-Contrast: Returning	Observable indicator: Learner's

📹 **VIDEO:** [Surface area of a box (cuboid)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	original phenomenon statement displayed on the board: 'The posho mill owner in Eldoret has a cylindrical drum and a rectangular metal box, both made from the same total area of iron sheet. The cylinder holds noticeably more maize.' Each learner silently writes a one-paragraph Final Prediction answering: 'Now that you have studied surface area and volume for seven lessons, write the most complete explanation you can for why the cylinder holds more maize. Include at least one formula and one numerical example.' Learners are reminded of their Lesson 1 prediction cards, which are returned to them at this point.	photograph and original driving question on the board. Distribute Lesson 1 prediction cards. Say: 'Before anyone speaks, I want you to read what you wrote on Day 1 — do not change it. Then write your final explanation in your exercise book. You have 8 minutes. Use formulae and numbers.' Set a visible timer. After 8 minutes, cold-call 3 learners (mix of confidence levels): 'Achieng, read us your Day-1 prediction.' Pause 10 seconds. 'Now read your final explanation.' Repeat for Omondi and Wanjiku. Ask the class: 'What is the biggest thing that changed in your thinking?' Wait 12 seconds before accepting responses. Record key shifts on the board in two columns: BEFORE | NOW.	the original prediction card forces metacognitive comparison. Learners must articulate conceptual change explicitly — they cannot simply state the correct answer; they must identify what was wrong or incomplete about their earlier thinking. This activates all prior lessons as a coherent body of evidence rather than isolated topics.	final written explanation includes (a) the correct formula for at least one solid, (b) a numerical comparison (e.g., two containers with equal surface area but different volumes), and (c) an explicit statement about why shape affects volume independently of surface area. Teacher scans written responses during the 8-minute silent period, noting any learner who still conflates surface area and volume — these learners are targeted for questioning during the Explain phase.
Observe Phase 📹 **VIDEO:** [Surface area of a box (cuboid)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group displays their home/extended project poster or physical model on a desk or wall section around the classroom. Posters must show: (i) chosen solid shape and labelled diagram with dimensions in cm or m; (ii) full working for total surface area calculation (material required); (iii) full working for volume/capacity calculation; (iv) a statement of community purpose (e.g., 'This cylindrical grain silo holds 2.8 m^3 of maize and requires 18.5 m^2 of iron sheet — enough for a 10-bag harvest store for a smallholder farmer in Uasin Gishu County'). Groups rotate every 6 minutes around all displays. Each visiting group leaves one sticky note: a GLOW (something mathematically strong) and one GROW (a question or suggestion). Learners record observations in their exercise books: for each design	Before the gallery walk, remind learners of the observation protocol: 'You are acting as the KEBS — Kenya Bureau of Standards — inspectors today. Your job is to verify the mathematics, not just admire the poster.' Circulate during rotations. Pause the class at the 12-minute mark and cold-call one group: 'Kamau's group — in 30 seconds, tell us your container shape, surface area, and volume.' After the full rotation (approximately 24 minutes for 4 rotations), bring the class back. Ask: 'Did any two groups use the same surface area but get different volumes? What does that tell us?' Wait 15 seconds. Cold-call 2 learners who left substantive GROW notes. Write the two most interesting design comparisons on the board for reference in the Explain phase.	Gallery Walk with Structured Observation: The gallery walk transforms individual projects into a shared data set. By recording multiple surface area–volume pairs from different solid shapes, learners build an empirical table in real time that mirrors the scientific practice of Analysing and Interpreting Data (NGSS SEP 4). The KEBS inspector framing raises the epistemic stakes — accuracy matters because real decisions (cost of materials, community food security) depend on correct calculations.	Observable indicator: Each learner's exercise book contains a table with at least 3 designs visited, each row showing shape, SA value, V value, and one observation. Teacher checks that sticky-note GLOWs and GROWs cite specific mathematical features (e.g., 'Your lateral surface area formula is correct but you forgot to add the base') rather than vague praise. Groups whose posters receive more than one GROW note about formula errors are flagged for teacher follow-up during the Explain phase.

	they visit, they note the shape, surface area value, volume value, and one thing they notice or question.			
Explain Phase 📹 **VIDEO:** Surface area of a box (cuboid) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group has 4 minutes at the front to present their design and respond to the GROW sticky notes left on their poster. They must answer at least one peer critique question mathematically — recalculating or justifying on the board if challenged. After all presentations, the teacher leads a whole-class Final Explanation discussion. Learners are asked to construct a single class consensus statement that resolves the driving question. They work in pairs for 3 minutes to draft a statement, then share. The class votes on the most complete statement and refines it collaboratively until every key mathematical idea is present: (1) formula for surface area does not contain volume; (2) for the same surface area, a cylinder encloses more volume than a rectangular prism of the same height because of the circle's isoperimetric efficiency; (3) the metalsmith's cylinder choice maximises grain storage per unit of iron sheet used. The consensus statement is written in full on the board and copied by all learners.	During group presentations, use the 'Press for Precision' protocol: after each group, ask one targeted question from the following bank — 'What units does your surface area have? What units does your volume have? Why are they different?' / 'If you doubled the height of your cylinder, what would happen to the volume? What would happen to the surface area? Would they change by the same factor?' / 'Your community design says it holds 500 kg of maize. How did you convert your volume in m^3 to a mass in kg?' After all 4 presentations (approximately 16 minutes total), say: 'Now let us answer the question that started this whole journey. In pairs, write one sentence that the posho mill owner in Eldoret could put on a sign in his mill to explain to customers why he chose a cylinder.' After 3 minutes, cold-call 4 pairs. Record key phrases on the board. Synthesise into one class statement, asking: 'What must we add to make this mathematically complete?' Ensure the final statement includes a formula reference. Allow 12 seconds of wait time before accepting each addition.	Press for Precision with Consensus Building (NGSS SEP 6 — Constructing Explanations): The structured critique requires learners to defend mathematical claims publicly, which surfaces and corrects residual errors. The consensus-statement protocol applies Constructing Explanations practice by demanding that the explanation be grounded in evidence (their own calculations), use mathematical models (formulae), and make a general claim (isoperimetric principle). Writing the statement on the board and copying it creates a permanent anchor for the final model.	Observable indicator: The class consensus statement on the board contains (a) the words 'surface area' and 'volume' used correctly and distinctly, (b) a reference to at least one formula (e.g., $V = \pi r^2 h$ for cylinder; $V = lwh$ for cuboid), and (c) a generalisation about shape efficiency. Teacher listens for any learner who uses 'bigger surface area means bigger volume' — this is the key misconception to extinguish. If it appears, ask: 'Can the class give me a counter-example from the gallery walk that disproves that?' Wait 10 seconds.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Surface area of a box (cuboid) Source: Khan	The full DQB — maintained across all 8 lessons — is displayed at the front. Each sticky-note question that was added over the unit is reviewed. The class is divided into 4 teams, each assigned a section of the DQB. Teams have 5 minutes to write a final answer	Before the DQB activity, read the original driving question aloud with theatrical emphasis: 'When Mwangi the metalsmith in Eldoret cut the same area of iron sheet to make two containers, one cylindrical and one rectangular — he may not have known it, but he	DQB Completion as Cumulative Sensemaking Audit (NGSS SEP 7 — Engaging in Argument from Evidence): Posting final answer cards against original questions makes the unit's knowledge-building arc visible and concrete. The thumbs audit is a whole-class	Observable indicator: All questions on the DQB have a response card. At least 80% of questions receive a thumbs-up from the class. For the main driving question, the posted answer references both surface area and volume formulae, names the specific shapes

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	card for each question in their section, citing the lesson and evidence that resolved it. Final answer cards are posted next to the original questions. The class then does a 'DQB Audit': the teacher reads one question, a team reads their answer, and the class gives a thumbs-up (fully resolved), thumbs-sideways (partially resolved), or thumbs-down (still uncertain). Questions still uncertain are noted as extension topics. The original driving question — 'Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?' — is addressed last, with the class consensus statement from Phase 3 read aloud as the final answer.	was doing mathematics. Today we will prove that he made the right choice with the cylinder, and we will explain exactly why.' Circulate during the 5-minute team writing period. During the DQB Audit, for any thumbs-sideways response, prompt: 'What evidence from which lesson would help us answer this more fully?' For the final driving question, ask the class to stand and read the consensus statement aloud together. Then say: 'This is what it means to think like a mathematician and an engineer. You have earned this answer.' Record which questions remain unresolved for follow-up in future lessons or as homework extension.	peer-review of explanations — learners must judge whether evidence is sufficient to resolve a question, which is a higher-order epistemic skill. Reading the final driving-question answer aloud together creates shared intellectual ownership of the explanation.	(cylinder and rectangular prism/cuboid), and makes a general design recommendation. Teacher notes any persistent thumbs-down questions and records learner names for targeted remediation.
Model Building Phase 📹 **VIDEO:** Surface area of a box (cuboid) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their Lesson 1 initial sketch model of the posho mill containers. Side-by-side with a clean page, they draw their Final Annotated Model. The final model must include: (a) labelled diagrams of a cylinder and a cuboid with dimensions; (b) surface area formulae for each: TSA(cylinder) = $2\pi r^2 + 2\pi rh$, TSA(cuboid) = $2(lb + bh + lh)$; (c) volume formulae: V(cylinder) = $\pi r^2 h$, V(cuboid) = lwh; (d) a worked numerical example showing two containers with equal surface area but different volumes; (e) an arrow labelled 'Isoperimetric Principle' pointing from the cylinder with the note 'Among solids with equal surface area, the cylinder encloses more volume than the rectangular box'; (f) a connection arrow to the phenomenon statement. Learners	Display a side-by-side comparison on the board: a deliberately naive 'Lesson 1 model' (just two rectangles labelled 'same size') next to a rich final model. Say: 'I drew what most people think on Day 1. Now show me what you actually know.' Allow 12 minutes for individual model drawing. Circulate and stamp or sign completed final models as 'KEBS Certified — Mathematically Sound Design.' During the last 8 minutes, facilitate a class share of Career Connections. Cold-call 5 learners to share one career connection each. Ask: 'Fatuma, which career did you choose and what specific calculation would they do?' After each response, ask the class: 'Can you think of a Kenyan example of someone in this field?' Close the lesson by returning to the	L1 vs Final Model Comparison (NGSS SEP 2 — Developing and Using Models): Placing the initial and final models side by side operationalises the concept of conceptual growth. Learners must identify specific additions — formulae, numerical examples, the isoperimetric principle — that they did not have on Day 1, making learning gains tangible. The Career Connections web applies NGSS SEP 8 (Obtaining, Evaluating, and Communicating Information) by requiring learners to connect abstract mathematical knowledge to real Kenyan professional contexts, consolidating the unit's relevance.	Observable indicator: Final model contains all 6 required elements listed in the learner experience. Career Connections web has at least 5 careers with specific Kenyan-context tasks. Teacher uses the 'KEBS Certified' stamp as a motivational indicator of completion — only models with all required elements receive the stamp, giving the teacher a rapid whole-class check. Learners without all elements are asked to complete the missing parts as the first task of the next lesson or as written homework.

	then individually complete a Career Connections web: they draw lines from 'Surface Area and Volume of Solids' to at least 5 careers, writing one specific task each career uses these concepts for (e.g., 'Jua kali metalsmith in Kamukunji, Nairobi — calculates iron sheet needed to fabricate a water tank'; 'NCWSC water engineer — designs cylindrical reservoir capacity for Nairobi's water supply'; 'KEBS standards officer — verifies that labelled container volumes match actual capacity').	phenomenon: 'Mwangi the metalsmith in Eldoret may not have known the words isoperimetric principle — but every time he hammers a cylinder instead of a box, he is using it. Now you can explain to him exactly why his instinct is mathematically correct.' Pause 10 seconds. 'That is what mathematics is for.'		

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) During the DQB Audit, how many questions received thumbs-down? What does this reveal about gaps in the unit's instructional sequence that should be addressed before the summative assessment? (2) Comparing Lesson 1 prediction cards with final explanations: which misconception persisted longest — the conflation of surface area with volume, or difficulty applying formulae for composite solids? Plan one targeted 20-minute remediation activity for this misconception before the unit test. (3) Which group's project design showed the most sophisticated mathematical reasoning? Could this design be used as a worked exemplar in future cohorts? (4) Were all learners able to articulate the isoperimetric principle in their own words, or was it only the highest-achieving learners? If the latter, consider whether an additional concrete activity (e.g., folding equal-perimeter shapes and measuring enclosed area) is needed at the start of the unit in future iterations. (5) Did the gallery walk surface any systematic calculation errors (e.g., using diameter instead of radius, omitting the base in total surface area)? Record these for inclusion in the review session. (6) Reflect on Kenya-context relevance: Did learners connect surface area and volume authentically to their community needs — water storage, grain storage, cooking — or did their projects feel formulaic? If the latter, consider starting the unit project briefing earlier (Lesson 3 or 4) to allow more iteration time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed each other's community container design projects displayed in a gallery walk around the classroom. They recorded surface area and volume values for cylindrical grain silos, rectangular water tanks, conical cooking-pot lids, and composite storage structures designed for Kenyan smallholder farming communities. They noted that different groups, working with similar amounts of material (surface area), produced containers with noticeably different holding capacities (volume), recreating the core puzzle of the Eldoret posho mill owner's cylindrical drum versus rectangular box at the scale of their own designed artefacts.
What did I learn?	Learners synthesised the full unit's content into a final annotated model: total surface area and volume formulae for cylinders ($TSA = 2\pi r^2 + 2\pi rh$; $V = \pi r^2 h$), cuboids ($TSA = 2(lb + bh + lh)$; $V = lwh$), cones, pyramids, frustums, spheres, and composite solids. They established that surface area measures the amount of material enclosing a solid while volume measures the space enclosed, and that these are independent properties determined by both size and shape. They articulated the isoperimetric principle — among solids with equal surface area, the cylinder encloses greater volume than a rectangular prism — as the mathematical resolution to the driving question. They connected these ideas to Kenyan careers including jua kali metalsmiths in Kamukunji, NCWSC water

	engineers, KEBS standards officers, and agricultural silo designers in the Rift Valley.
How does this explain the phenomenon?	Using the class consensus statement constructed during the Explain phase and the completed Driving Question Board, learners produced a full written explanation resolving the anchoring phenomenon: the Eldoret posho mill owner's cylindrical drum holds more maize than the rectangular metal box because, for the same total surface area of iron sheet, the cylindrical shape encloses a greater volume. This is because the circle — the cross-section of the cylinder — is the most area-efficient closed curve for a given perimeter (isoperimetric principle), meaning the circular base of the cylinder packs more capacity per unit of bounding material than the rectangular base of the box. Mathematically, two containers with equal TSA but different shapes will have different volumes, and the cylinder consistently wins this comparison against a same-height rectangular prism. A Kenyan artisan who understands surface area and volume formulae can therefore calculate the optimal solid shape and dimensions to maximise storage capacity for a given budget of metal sheet — turning an intuitive craft skill into a precise, reproducible mathematical design process.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.